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Scientific-Analytical Teyf Group

Probability Theory for Data Science and Machine Learning

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November 2024

Chapter 6: Probability for Data Science and Machine Learning

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Understanding Probability

Probability theory is nothing but common sense reduced to **calculation**. — Pierre Laplace,1812



It is wonderful that this science was invented in the beginning to investigate games of chance, but today it should become the most important human knowledge.

- Common sense refers to our **intuitive understanding** of likelihood and uncertainty.
- **Question 1**: what is probability?
- Our world is full of **uncertainties** we want to **measure**.
- **Probability theory** is a mathematical framework for **representing uncertain statements**. It provides a means of **quantifying(measuring) uncertainty** and **axioms for deriving new uncertain statements**.
- Probability theory is a fundamental tool of many disciplines of science and engineering.

Understanding Probability

- While probability theory allows us to **make uncertain statements** and **reason** in the presence of uncertainty, **information** allows us to quantify the **amount of uncertainty** in a probability distribution.

- Researchers have made compelling arguments for quantifying uncertainty using probability since at least the 1980s.

- Nearly all activities require some ability to reason in the presence of uncertainty. In fact, beyond mathematical statements that are **true by definition**, it is difficult to think of any proposition that is absolutely true or any event that is absolutely guaranteed to occur.

- There are three possible sources of uncertainty:

  1. **Inherent stochasticity** in the system being modeled.

  2. **Incomplete observability**: Even **deterministic systems** can appear stochastic when we cannot observe all of the variables that drive the behavior of the system.

  3. **Incomplete modeling**: When we use a model that must discard some of the information we have observed, the discarded information results in uncertainty in the model's predictions.

Understanding Probability

- When constructing **automated reasoning systems**, **classical Boolean logic** does not allow us to express certain forms of **plausible reasoning**.
- Probability theory can be seen as the **extension of Boolean logic** to deal with uncertainty.
- **Logic** provides **a set of formal rules** for determining what propositions are implied to be true or false given the assumption that some other set of propositions is true or false.
- Probability theory provides **a set of formal rules** for determining the **likelihood of a proposition** being true given the likelihood of other propositions.
- **Question**: Thus, how do we measure something that we are uncertain about?
- Alternative Method: **Fuzzy Logic**

❖ **Boolean Logic vs. Fuzzy Logic vs. Probability**

- Boolean logic, fuzzy logic, and probability are different approaches to **handling uncertainty and reasoning**, each with its own strengths and applications.

Understanding Probability

❖ Boolean Logic vs. Fuzzy Logic vs. Probability

- **Boolean Logic:**
  - Boolean logic deals with binary truth values: true (1) or false (0).
  - Example:
    - Statement: "It is raining."
    - Boolean value: True (1) or False (0)

  - This approach is precise but lacks nuance for real-world scenarios where things aren't always black and white.

- **Fuzzy Logic:**
  - Fuzzy logic allows for **degrees of truth**, representing partial truths between 0 and 1, where 0 represents absolute falsity and 1 represents absolute truth.

  - Fuzzy logic, developed by Lotfi A. Zadeh, is an extension of Boolean logic that allows for degrees of truth rather than binary true/false values.

  - Example:
    - Statement: "The room is warm."
    - Fuzzy value: 0.7 (on a scale of 0 to 1, where 0 is cold and 1 is hot)

Understanding Probability

❖ Boolean Logic vs. Fuzzy Logic vs. Probability Theory

- **Probability:**
  - Probability quantifies the likelihood of an event occurring, ranging from 0 (impossible) to 1 (certain).
  - Probability is how strongly we **believe** an event will happen, often expressed as a percentage.
  - Example:
    - Statement: "It will rain tomorrow."
    - Probability value: 0.75 (75% chance of rain)

- In **telecom industry**, for example:
  - Boolean logic for simple decision-making, like **network congestion detection**.
  - Fuzzy logic for modeling uncertainty and nuance, like **network quality assessment**.
  - Probability for analyzing and modeling chance events, like **network failure prediction**.

- Alternatively, probability can be expressed as an **odds** $P(x)$:

$$O(x) = \frac{P(x)}{1 - P(x)}, \qquad O(x) = \frac{0.75}{1 - 0.75} = 3$$

Understanding Probability

- While many people feel more comfortable expressing probabilities as percentages or proportions, **odds** can be a helpful tool.

- If we have an odds of 3.0, that means we **feel** an event is three times more likely to happen than not to happen. That can be **more intuitive** to **describe a belief** than a percentage. For this reason, odds are helpful **quantifying subjective beliefs**.

❖ **Likelihood vs. Probability**

  - **Likelihood** is similar to probability, and it is easy to confuse the two (many dictionaries do as well).

  - You can get away with using "probability" and "likelihood" interchangeably in everyday conversation. However, we should pin down these differences.

  - Probability is about quantifying predictions of events yet to happen, whereas likelihood is measuring the frequency of events that already occurred.

  - In statistics and machine learning, we often use likelihood (the past) in the form of data to predict probability (the future).

Understanding Probability

❖ Likelihood vs. Probability

- Likelihood is a specific concept used within **statistical inference**.
- Likelihood functions are used to **estimate parameters of statistical models**.
- Maximum Likelihood Estimation, a common method for parameter estimation.
- In essence, likelihood is a tool that can be used within inferential statistics, but inferential statistics is a much broader field that encompasses many other concepts and techniques beyond just likelihood.
- Likelihood plays a crucial role in many inferential methods, particularly in **parameter estimation** and **model selection**, but it is just one component of the larger framework of **inferential statistics**.

- **Question**: What is Statistics?

- **Question**: What is statistical inference/ inferential statistics ?

Understanding Probability

- In many cases, it is more practical to use a **simple but uncertain rule** rather than a **complex but certain one**, even if the true rule is **deterministic** and our modeling system has the fidelity to accommodate a complex rule.

- For example, the simple rule "Most birds fly" is cheap to develop and is broadly useful, while a rule of the form, "Birds fly, except for very young birds that have not yet learned to fly, sick or injured birds that have lost the ability to fly, flightless species of birds including the cassowary, ostrich and kiwi. . . " is expensive to develop, maintain and communicate, and after all of this effort is still very brittle and prone to failure.

- Given that we need a means of representing and reasoning about uncertainty, it is not immediately obvious that probability theory can provide all of the tools we want for artificial intelligence applications.

- Probability theory was originally developed to analyze the frequencies of events(**repeatable** experiments).

- **Question:** What about **non-repeatable** experiments?

- If a doctor analyzes a patient and says that the patient has a 40% chance of having the flu, this means something very different.
- In the case of the doctor diagnosing the patient, we use probability to **represent** a ***degree of belief***, with 1 indicating absolute certainty that the patient has the flu and 0 indicating absolute certainty that the patient does not have the flu.

Understanding Probability

- **Question 2**: we are all familiar with the phrase "the probability that a coin will land heads is 0.5". But what does this mean?

❖ **Two Different Interpretations of Probability**:

- There are actually at least two different interpretations of probability. One is called the **frequentist** interpretation.
- In this view, probabilities represent long run frequencies of events.
- For example, the above statement means that, if we flip the coin many times, we expect it to land heads about half the time.
- The other interpretation is called the **Bayesian interpretation of probability**. In this view, **probability is used to quantify our uncertainty about something**; hence it is fundamentally related to information rather than repeated trials.
- In the Bayesian view, the above statement means we **believe** the coin is equally likely to land heads or tails on the next toss.

Understanding Probability

❖ **Two Different Interpretations of Probability**:

- One big advantage of the Bayesian interpretation is that it can be used to model our uncertainty about events that do not have long term frequencies.

- For example, we might want to compute the probability that the polar ice cap will melt by 2030. This event will happen zero or one times, but cannot happen repeatedly. Nevertheless, we ought to be able to quantify our uncertainty about this event; based on how probable we think this event is, we will (hopefully!) take appropriate actions.

- To give some more machine learning oriented examples, we might have received a specific email message, and want to compute the probability it is spam. Or we might have observed a "blip" on our radar screen, and want to compute the probability distribution over the location of the corresponding target (be it a bird, plane, or missile). In all these cases, **the idea of repeated trials** does not make sense, but the **Bayesian interpretation** is valid and indeed quite natural.

- **Summary**: Probability can be thought of as the fraction of times an event occurs(frequentist interpretation), or as a ***degree of belief*** about an event(Bayesian interpretation/ **subjective probability**).

- Fortunately, the basic rules of probability theory are the same, no matter which interpretation is adopted.

Understanding Probability

❖ Toy Models

Given so many exciting applications, you may wonder why we will spend so much time thinking **about toy models** like coins and dice. By understanding these thoroughly we will develop a good feel for the simple essence inside many complex real-world problems. In fact, the modest coin is a realistic model for any situations with two possible outcomes: success or failure of a treatment, an airplane engine.

Understanding Probability

❖ Probability vs. Statistics

- Sometimes people use the terms *probability* and *statistics* interchangeably, and while it is understandable to conflate the two disciplines, they do have distinctions.
- Statistics is applied probability(in statistics we apply probability to draw conclusions from data) ,and probability is the mathematical description of random events.
- While probability is a **deductive** science—starting from definitions we can prove mathematical theorems such as the **law of large numbers**—statistics is as much art as science.
- Statistics uses probability theory to describe and draw **inferences** from data.
- Probability theory was a purely mathematical theory in which we obtained the probability of other events based on a basic principle and mathematical inferences from a given model (the probability of a series of events). Models are assumed and we have nothing to do with whether they correspond to reality or not.
- Probability: uncertain world and perfect knowledge of the uncertainty
- Statistics: data in an uncertain world and imperfect knowledge of the uncertainty

Understanding Probability

❖ Probability vs. Statistics

- *Probability* is purely theoretical of how likely an event is to happen and does not require data. *Statistics*, on the other hand, cannot exist without data and uses it to discover probability and provides tools to describe data.

- **Probability example**

  1) Think of predicting the outcome of rolling a 4 on a die (that's the singular of dice). Approaching the problem with a pure probability mindset, one simply says there are six sides on a die. We assume each side is equally likely, so the probability of getting a 4 is 1/6, or 16.666%.

  2) You have a fair coin (equal probability of heads or tails). You will toss it 100 times. What is the probability of 60 or more heads? There is only one answer (about 0.028444) and we will learn how to compute it.

Understanding Probability

❖ Probability vs. Statistics

▪ Statistics example

1) However, a zealous statistician might say, "No! We need to roll the die to get data. If we can get 30 rolls or more, and the more rolls we do the better, only then will we have data to determine the probability of getting a 4." This approach may seem silly if we assume the die is fair, but what if it's not? If that's the case, collecting data is the only way to discover the probability of rolling a 4. We will talk about **hypothesis testing** in statistics chapter.

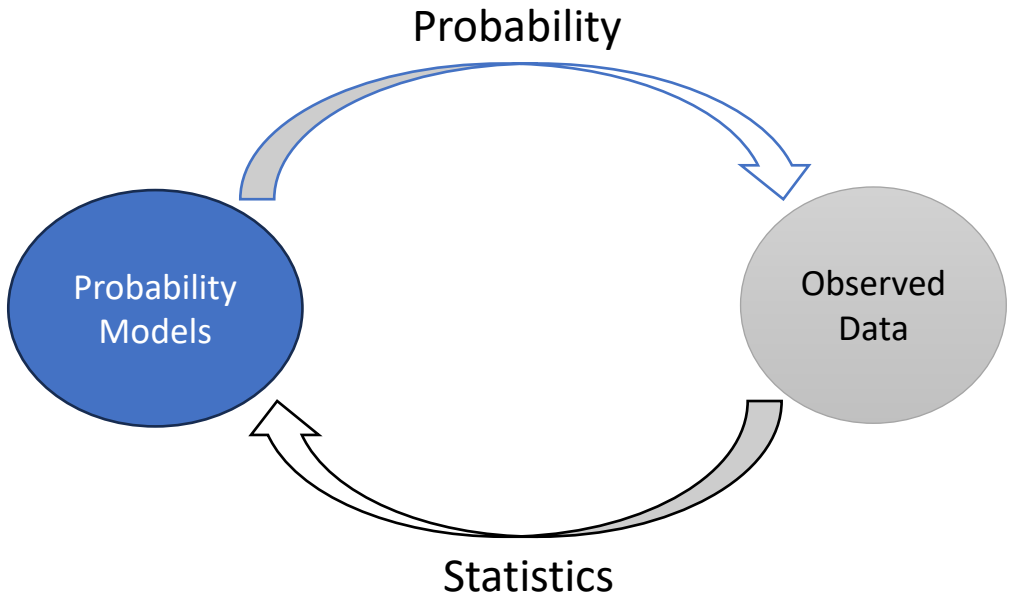
2) You have a coin of unknown provenance. To investigate whether it is fair you toss it 100 times and count the number of heads. Let's say you count 60 heads. Your job as a statistician is to draw a conclusion (inference) from this data. There are many ways to proceed, both in terms of the form the conclusion takes and the probability computations used to justify the conclusion. In fact, different statisticians might draw different conclusions.

- Note that in the probability example the random process is fully known (probability of heads = 0.5). The objective is to find the probability of a certain outcome (at least 60 heads) arising from the random process. In the statistics example, the outcome is known (60 heads) and the objective is to illuminate the unknown random process (the probability of heads).

Understanding Probability

❖ Probability vs. Statistics

- Probability laws describe the uncertainty in the data we might observe.
- Statistics describe the salient features of the data we do observe, and allow us to draw conclusions(inferences) about which probability model actually reflects the true state of nature



Statistical Data Processing in a nutshell

Step 1: Postulate a probability model(or models) that can be expected to reasonably capture the uncertainties in the data.

Step 2: Collect data

Step 3: Formulate **statistics** that allow us to interpret or understand our probability models.

Understanding Probability

❖ Uncertainty in Artificial Intelligence and Dat Science

- In artificial intelligence applications, we use probability theory in two major ways. First, the laws of probability tell us how AI systems should reason, so we design our algorithms to compute or approximate various expressions derived using probability theory. Second, we can use probability and statistics to theoretically analyze the behavior of proposed AI systems.

- Many branches of computer science deal mostly with entities that are entirely deterministic and certain.

- Given that many computer scientists and software engineers work in a relatively clean and certain environment, it can be surprising that machine learning makes heavy use of probability theory.

- For software engineers and developers, computers are **deterministic**. You write a program, and the computer does what you say.

- Three main sources of uncertainty in machine learning:
  1. uncertainty in the data(finite size of datasets, noise and bias)
  2. uncertainty in the machine learning model,
  3. uncertainty in the predictions produced by the model.

Understanding Probability

❖ Uncertainty in Artificial Intelligence and Dat Science

- Applied machine learning requires getting comfortable with uncertainty. Uncertainty means working with imperfect or incomplete information. Uncertainty is fundamental to the field of machine learning, yet it is one of the aspects that causes the most difficulty, especially those coming from a developer background.

- **Predictive modeling** with machine learning involves fitting a model to **map** examples of inputs to an output, such as a number in the case of a **regression** problem or a class label in the case of a **classification** problem. Naturally, you may ask reasonable questions, such as:

  - What are the best features that I should use?

  - What is the best algorithm for my dataset?

- The answers to these questions are unknown and might even be unknowable, at least exactly.

- This is the major cause of difficulty. The reason that the answers are unknown is because of uncertainty, and the solution is to systematically evaluate different solutions until a good or good-enough set of features and/or algorithm is discovered for a specific problem.

Understanding Probability

❖ Uncertainty in Artificial Intelligence and Dat Science

- **Noise in Observations:**
  - The real world, and in turn, real data, is messy or imperfect. As practitioners, we must remain skeptical of the data and develop systems to expect and even harness this uncertainty. This is why so much time is spent on reviewing statistics of data and creating visualizations to help identify those aberrant or unusual cases: so-called **data cleaning**.

- **Incomplete Coverage of the Domain:**
  - Observations from a domain used to **train a model** are a **sample** and incomplete by definition. In statistics, a **random sample** refers to a collection of observations chosen from the domain without systematic **bias** (e.g. uniformly random). Nevertheless, there will always be some limitation that will introduce bias.

  - We aim to collect or obtain a suitably **representative** random sample to train and evaluate a machine learning model. Often, we have little control over the sampling process.

  - No matter how well we guide our models to generalize, we can only hope that we can cover the cases in the **training dataset** and the salient cases that are not.

Understanding Probability

❖ Uncertainty in Artificial Intelligence and Dat Science

- **Incomplete Coverage of the Domain:**
  - This is why we split a dataset into train and test sets or use resampling methods like ***k*-fold cross-validation**. We do this to handle the uncertainty in the representativeness of our dataset and estimate the performance of a modeling procedure on data not used in that procedure.

- **Imperfect Model of the Problem**
  - A machine learning model will always have some error. This is often summarized as *all models are wrong*, or more completely in an aphorism by George Box:

All models are wrong but some are useful.

  - This does not apply just to the model, the artifact, but the whole procedure used to prepare it, including the choice and preparation of data, choice of **hyperparameters**, and the interpretation of model predictions.

  - Model error could mean imperfect predictions. This type of error in prediction is expected given the uncertainty we have about the data that we have just discussed, both in terms of noise in the observations and incomplete coverage of the domain.

Understanding Probability

❖ Uncertainty in Artificial Intelligence and Dat Science

▪ Imperfect Model of the Problem

- Another type of error is an error of omission. We leave out details or **abstract** them in order to **generalize** to new cases. This is achieved by selecting models that are simpler but more robust to the specifics of the data, as opposed to complex models that may be highly specialized to the training data. As such, we might and often do choose a model known to make errors on the training dataset with the expectation that the model will generalize better to new cases and have better overall performance.

- In many cases, it is more practical to use a simple but uncertain rule rather than a complex but certain one, even if the true rule is deterministic and our modeling system has the fidelity to accommodate a complex rule.

- Nevertheless, predictions are required. Given we know that the models will make errors, we handle this uncertainty by seeking a model that is good enough.

Understanding Probability

❖ How to Manage Uncertainty

- Uncertainty in applied machine learning is managed using probability theory. Probability is the field of mathematics designed to quantify, handle, manipulate, and harness uncertainty.

- Probability theory provides a consistent framework for the quantification and manipulation of uncertainty and forms one of the central foundations for **pattern recognition**.

- In fact, probability theory is central to the broader field of artificial intelligence.

- Agents can handle uncertainty by using the methods of probability and decision theory, but first they must **learn** their probabilistic theories of the world from **experience**. In essence, this statement encapsulates a fundamental principle in artificial intelligence and cognitive science: effective decision-making under uncertainty requires not just probabilistic reasoning tools, but also the ability to learn and refine probabilistic models of the world through experience.

- **In terms of noisy observations**, probability and statistics help us to understand and quantify the expected value and variability of variables in our observations from the domain.

Understanding Probability

❖ How to Manage Uncertainty

- **In terms of the incomplete coverage of the domain**, probability helps to understand and quantify the expected distribution and density of observations in the domain.
- **In terms of model error**, probability helps to understand and quantify the expected capability and variance in performance of our predictive models when applied to new data.
- But this is just the beginning, as probability provides the foundation for the iterative training of many machine learning models, called maximum likelihood estimation, behind models such as linear regression, logistic regression, artificial neural networks, and much more.
- Probability also provides the basis for developing specific algorithms, such as Naive Bayes, as well as entire subfields of study in machine learning, such as graphical models like the Bayesian Belief Network.
- In summary: many machine learning models are trained using an iterative algorithm designed under a probabilistic framework.

Understanding Probability

❖ Why Learn Probability for Machine Learning

1. Class Membership Requires Predicting a Probability
2. Some Algorithms Are Designed Using Probability
3. Models Are Trained Using a Probabilistic Framework
4. Models Can Be Tuned With a Probabilistic Framework
5. Probabilistic Measures Are Used to Evaluate Model Skill

Understanding Probability

❖ Definitions

- Despite the wide application of probability and despite the fact that we constantly use such concepts in our daily life, **there is no single scientific definition for probability**, and throughout the history of the development of probability theory, there have been various definitions of probability, each of which has been criticized by others.

Definition 1. The classical definition of probability (by Pascal in the 17th century):

$$P(A) = \frac{N_A}{N}$$

Definition 2. Definition of relative frequency (statistical definition):

- If we do the **random experiment** *n* times, we have for large n:

$$P(A) = \lim_{n \to \infty} \frac{n_A}{n}$$

Understanding Probability

❖ Definitions

Definition 3. Subjective Definition:
The attitude towards probability is a measure of the ***degree of belief*** in a matter.

- The previous definitions have problems that cannot be used as the basis of a **mathematical theory.**

Definition 4. Axiomatic Definition:
Probability is a **set function** P(A) that assigns to every event A a number called the "probability of A" such that:

Axiom 1. The probability of an event is greater than or equal to zero

$$P(A) \geq 0$$

Axiom 2. The probability of sample space is one

$$P(\Omega) = 1$$

Axiom 3. If $A \cap B = \emptyset$ **:** $P(A \cup B) = P(A) + P(B)$

Understanding Probability

❖ Definitions

- This definition was proposed by Kolmogorov in 1933 (though it took many years to gain attention).
- Here, the probability is presented based on the theory of measure and each event is assigned to a number(which is called the **probability of that event** and must apply to the principles of the triple axioms).
- What number is assigned to each event (assuming the satisfaction of the axioms) is arbitrary and may not be in full agreement with reality. But **assuming** the truth of these assumed probabilities for events, we can obtain the probability of other desired events by using **probability theory**.
- **Probabilistic Modeling**:
  1. We are faced with processes that, due to their complexity, we want to model the occurrence of events in a probabilistic manner (such as tossing a coin or rolling a dice or weather forecasting or network failure prediction). We consider $P(A_i)$ for event A_i .
  2. Assuming that $P(A_i)$ meet the principles of the triple axioms, we calculate $P(B_i)$ by the probability theory(**Objective**).
  3. We predict that event B_i will happen with probability $P(B_i)$ (**Subjective**).

Understanding Probability

❖ Probabilistic Modeling

- Steps 1 and 3 deal with the outside world, while the second step is purely conceptual, where we deal with the probabilistic model we have built of the outside world. In the second stage, there is no doubt.

- We use the axiomatic definition of probability. Of course, in steps 1 and 3, which is to attribute probabilities to the outside world, it is useful to use other definitions of probability.

- It may be said that the results will not necessarily match the actual results. But it is the same in any other science.

- We **assume** and analyze models that are **often much simpler than the outside world**, and we **predict** the result of the analysis by **approximating** what happens in the outside world. See: No Free Lunch Theorem, Model.

- For example, in circuit theory: $R = \frac{V}{I}$. The **actual** resistance is different from what we have modeled in our **ideal analysis**, and therefore the circuit response we have in the outside world does not completely match the response given by the circuit theory. But it can be used as an **approximation**.

Understanding Probability

❖ Probabilistic Modeling

- Another point is that it may be thought that the use of probabilities is always due to **our ignorance** of the phenomena and laws that govern them, and therefore we proceed with probabilistic analysis/modeling.

- But in fact, it should be known that in many cases when we use **deterministic analysis**, we have simplified the problem so much that the amount of ignorance and discarding information is much higher.

- The resistance value of 3Ω is not exactly 3Ω. Rather, it has a **distribution** around 3Ω.

- When we connect two resistors of 3Ω in series, if we only use the **averages**, we say that the resulting resistance is 6Ω. While it also has a certain **range of values** and **distribution.**

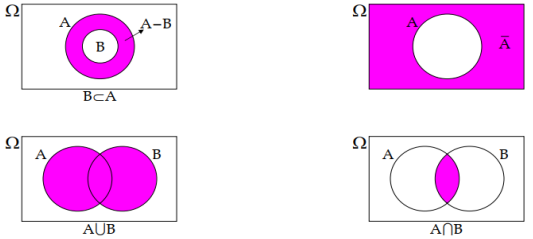
- Or when the ball is thrown, in simple deterministic analysis, **a point** is predicted for the place where it hits the ground, while in **probabilistic analysis**, an area is obtained by determining the **probability density of different points.**

- Probabilistic models provide **richer insights** and more **robust decision-making** tools compared to deterministic models.

Understanding Probability

❖ Definitions

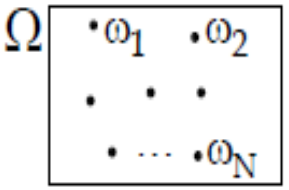
- A **set function** P(A) is a function that:
  1. Takes **events** (which are sets) as its input.
  2. Assigns a real number (the probability) to each event.



- This set function allows us to **assign** probabilities to various **events** (**subsets** from the **sample space**) within a probability space, providing a **mathematical framework** for **quantifying uncertainty.** In this sense, we need **set theory**.

- A **sample space** is the **set** of all possible outcomes of a **random experiment.** Note that an experiment may be performed only once, but it is random.

- A valid sample space must be:
  1. Mutually exclusive: Outcomes cannot overlap.
  2. Collectively exhaustive: All possible outcomes must be included.
  3. Have the right granularity: Outcomes should be at an appropriate level of detail.



- **Examples of sample space:**
  1. Tossing a coin: $\Omega = \{H, T\}$
  2. Rolling a dice: $\Omega = \{1, 2, 3, 4, 5, 6\}$
  3. The lifespan of an electrical element in the circuit: $\Omega = \{t:\ 0 \leq t < +\infty\}$

Understanding Probability

- To determine the sample space:
  1. Define the experiment clearly.
  2. List all possible outcomes.
  3. Ensure each outcome is unique.

- **Examples of event:**
  1. $A = \{2, 4, 6\}$
  2. $A = \{t: t \geq 10\}$

- **Understanding sample spaces** is crucial for **modeling** and **interpreting** data in statistics and probability theory.

- A **random experiment** is an experiment whose outcome is not known in advance, such as tossing a dice or a coin.

- **In summary**, the **probability space** or **probability model** of a random experiment is made up of the following factors:
  1. Set Ω contains all possible results.
  2. Subsets of Ω, which are called events.
  3. P(A) is attributed to each of the events according to **principles of the triple axioms.**

Previous Sections:
Verbal Description of Probability

Translating into



Next Sections:
Mathematics Symbols(Language)

❖ **In the following sections, be sure to pay attention to**

- whether you are dealing with an event or a random variable.
- whether you are dealing with one random variable or two random variables or sequence of random variables.
- whether you are dealing with the random variable itself or with its value (realization of the random variable) in formulas.
- Different concepts will be introduced many times and then they will be translated into symbols.
- whether you are dealing with a random variable or random sample.

Probability Math

- Now that we understand the **sample space** and the **event** the **third part** of the model is to attribute the probability to the events.

- Each event A is attributed to the number $P(A)$ so that it meets the **Kolmogorov axioms**.

- Now using these axioms we can derive the probability of different events from the probability of given events.

  - **Theorem 1**. $P(\bar{A}) = 1 - P(A)$

    Proof:

$$\Omega = A \cup \bar{A}$$

$$P(\Omega) = P(A \cup \bar{A})$$

Axiom 2 ↓ ↓ Axiom 3

$$1 = P(A) + P(\bar{A}) \Rightarrow P(\bar{A}) = 1 - P(A)$$

  - **Theorem 2**. $P(A) \leq 1$

    Proof: Theorem 1: $P(A) = 1 - P(\bar{A})$

    Axiom 1: $P(\bar{A}) \geq 0$

$$P(A) \leq 1 \overset{Axiom\ 1}{\rightarrow} 0 \leq P(A) \leq 1$$

Probability Math

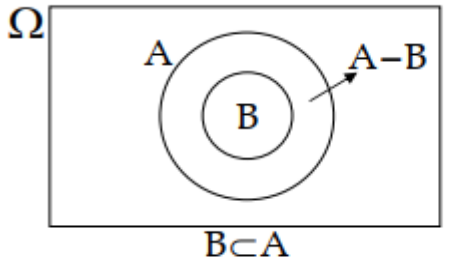
- **Theorem 3**. If $B \subset A$ *then*: $P(A-B) = P(A) - P(B)$

  Proof:

$$A = B \cup (A - B)$$

$$\downarrow \text{Axiom 3}$$

$$P(A) = P(B) + P(A-B) \Rightarrow P(A-B) = P(B) - P(A)$$



- **Theorem 4**. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

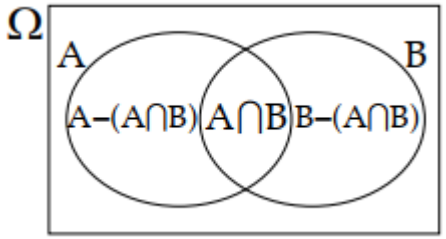
  Proof:

$$A \cup B = (B - (A \cap B)) \cup (A \cap B) \cup (A - (A \cap B)) \quad \downarrow \text{Axiom 3}$$

$$P(A \cup B) = P(B - (A \cap B)) + P(A \cap B) + P(A - (A \cap B)) \quad \downarrow \text{Theorem 3}$$

$$= P(B) - P(A \cap B) + P(A \cap B) + P(A) - P(A \cap B)$$

$$= P(B) + P(A) - P(A \cap B)$$



Probability Math

❖ Determining the probability of events

- To model a random experiment, we need to determine the probability of **all events** (A set of n members has 2^n subsets).
- But according to **the axioms and theorems**, there is no need to attribute probability to every event.
- For example, if we determine P(A), then $P(\bar{A})$ will be known by itself($P(\bar{A}) = 1 - P(A)$). Therefore, by determining the probability of a minimum number of events, the probability of all events will be known.
- If the number of elements Ω is **uncountable,** for example, the sample space of the start time of a telephone conversation or the time of an electronic element's breakdown, here every interval $\{t_1 \leq t \leq t_2\}$ is an event and $\Omega = \{t\colon\ 0 \leq t < +\infty\}$.
- Often, in such cases, the probability of the simple event $P\{t = t_i\}$ is zero, although Ω is the set of these simple events.
- Here, the **probability space** can not be determined by the probability of simple events. Instead, the **probability of intervals** should be determined. For this purpose, we introduce a function that expresses the **probability density**.

Probability Math

❖ Determining the probability of events

- **Probability Density Function**:

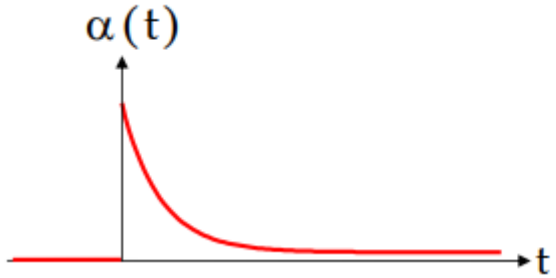
$$P\{t_1 \le t \le t_1 + dt\} = \alpha(t_1)dt$$

$$P\{t_1 \le t \le t_2\} = \int_{t_1}^{t_2} \alpha(t)dt$$

- Since the probability of any interval cannot be negative, then: $\alpha(t) \geq 0$ and since the probability Ω of is equal to one:

$$\int_{\Omega} \alpha(t)dt = 1$$

- In the example above: $\Omega = \{t: 0 \leq t < +\infty\}$



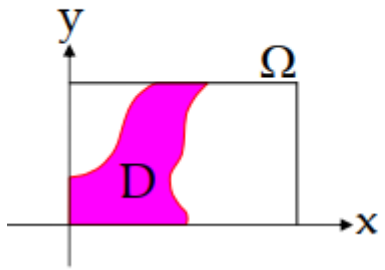
$$\int_0^{+\infty} \alpha(t)dt = 1$$

Probability Math

❖ Determining the probability of events

- **Probability Density Function**:
  - Since all the events of Ω can be expressed as the union and intersection of intervals, the probability of all events is determined.
  - Similarly, we have in two-dimensional space:

$$P\{(x,y)\in D\}=\iint_D \alpha(x,y)\,dx\,dy$$



$$\alpha(x,y)\geq 0 \quad , \quad \iint_\Omega \alpha(x,y)\,dx\,dy=1$$

- We calculate the probability based on the axiomatic definition, but we use **other definitions** according to their problems, not as a definition, but in steps 1 and 3 of modeling to establish a connection between the model and the outside world.

Probability Math

❖ Determining the probability of events

- The **classical definition** is no longer true if the results of a random experiment are not **equally likely**.
- We usually express equal probability simply because there is no reason for preference, which is true in the case of coins or dice, but for example, it is not true in the case of the birth of a child.
- In this way, we use the classical definition not as a definition, but in cases where equal probability is reasonable, to choose $P(A_i)s$.
- **The principle of insufficient reason:**

> If a random experiment has N possible outcomes and we have no information about how they will occur, we **must assume** that they are equally likely, i.e.
>
> $$P(\omega_1) = P(\omega_2) = \ldots = P(\omega_N) = \frac{1}{N}$$

Probability Math

- **Conditional Probability**
  - If M is an event in Ω that $P(M) \neq 0$:

$$P(A \mid M) \triangleq \frac{P(A \cap M)}{P(M)}$$

  - Empirical **interpretation** (relative frequency): If a random experiment is performed n times, we have

$$\frac{P(A \cap M)}{P(M)} = \frac{\frac{n_{A \cap M}}{n}}{\frac{n_M}{n}} = \frac{n_{A \cap M}}{n_M}$$

- **Total Probability Theorem**
  - If A_i are disjoint (**mutually exclusive**) events and $P(\bigcup_{i=1}^{m} A_i) = 1$, then for any arbitrary event B from Ω we have:

$$P(B) = \sum_{i=1}^{m} P(B|A_i)P(A_i)$$



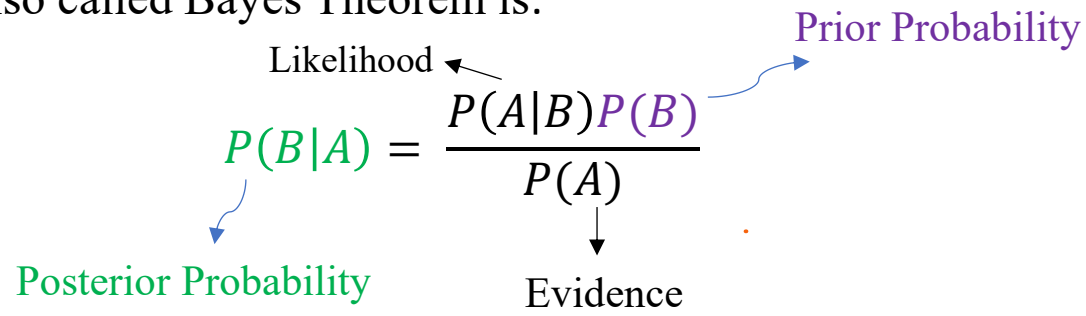
$$\forall i \neq j: A_i \cap A_j = \emptyset \quad , \quad \bigcup_{i=1}^{m} A_i = \Omega$$

  - **Question:** What is difference between = and $\triangleq$?

Probability Math

❑ Bayes Theorem

- Sometimes we have $P(A|B)$ and want $P(B|A)$.
- Bayes rule, also called Bayes Theorem is:



- **Posterior** refers to probability obtained after evidence(data) has been taken into consideration.
- **Prior** refers to probability before evidence(data) has been taken into consideration.

▪ Example: Medical Diagnosis

- The **base rate fallacy** is one of many examples showing that it's easy to confuse the meaning of $P(B|A)$ and $P\ (A|B)$ when a situation is described in words. This is one of the key examples from probability and it informs much of our practice and interpretation of statistics. You should strive to understand it thoroughly.

Probability Math

- **Example: Medical Diagnosis**
  - Consider a routine screening test for a disease. If the test is positive, what is the probability you have disease?
    - We'll use the following notation for the relevant events:
      - D^+ = 'you have the disease'
      - D^- = 'you do not have the disease
      - T^+ = 'you tested positive'
      - T^-= 'you tested negative'.

    - That obviously depends on how reliable the test is.
    - Suppose you are told the test has a **sensitivity** of 80%, which means, if you have cancer, the test will be positive with probability 0.8. In other words,

$$P(T^+|D^+) = 0.8$$

    - Many people conclude they are therefore 80% likely to have cancer. But this is false! It ignores the **prior probability** of having cancer. Frequency of the disease in the population (**base rate**) is 0.4%, which fortunately is quite low

$$P(D^+) = 0.004$$

    - Ignoring this prior is called the **base rate fallacy**.

Probability Math

- **Example: Medical Diagnosis**
  - We also need to take into account the fact that the test may be a **false positive** or **false alarm**. Unfortunately, such false positives are quite likely:

$$P(T^+|D^-) = 0.1$$

  - Using these three terms, Bayes rule and total probability theorem we can compute the correct answer as follows:

$$P(D^+|T^+) = \frac{P(T^+|D^+)P(D^+)}{P(T^+)} = \frac{P(T^+|D^+)P(D^+)}{P(T^+|D^+)P(D^+)+P(T^+|D^-)P(D^-)}$$

$$= \frac{0.8 \times 0.004}{0.8 \times 0.004 + 0.1 \times (1 - 0.004)} = 0.031$$

  - Thus, if your is test positive, you only have about a 3% chance of actually having cancer.
  - Probability theory can be used to represent and updates our **beliefs** about the state of the world.
  - These numbers are from (McGrayne 2011, p257). Based on this **analysis**, the US government decided not to recommend annual test.

Probability Math

- It and similar examples are at the heart of many statistical misunderstandings.
- Bayes' rule is a pillar of probability and statistics.
- We have seen that Bayes' rule allows us to 'invert' conditional probabilities. The art of **statistical inference** involves deciding how to proceed when one (or more) of the terms on the right side of Bayes' rule is unknown.

- **Example: Telecom Fraud Detection**

Scenario:

Suppose a telecom company wants to detect fraudulent activities in its network. They have historical data indicating the likelihood of certain behaviors being fraudulent.

Definitions:

- **Event (x = 1)**: A specific behavior (e.g., making an unusually high number of international calls in a short period).
- **Event (y = 1)**: The account is fraudulent.

Given Data:

- **Sensitivity (True Positive Rate)**: ($p(x = 1 \mid y = 1) = 0.9$) (If the account is fraudulent, the behavior is observed with 90% probability).
- **Prior Probability of Fraud**: ($p(y = 1) = 0.005$) (The overall probability of an account being fraudulent is 0.5%).
- **False Positive Rate**: ($p(x = 1 \mid y = 0) = 0.02$) (If the account is not fraudulent, the behavior is observed with 2% probability).

Probability Math

- **Calculation Using Bayes' Theorem:**
  - To find the probability that an account is fraudulent given the observed behavior, we use Bayes' theorem: …
    (**solution**: $P(y=1|x=1) \approx 0.184$)
- **Interpretation:**
  - If the specific behavior is observed, there is approximately an 18.4% chance that the account is fraudulent. This is significantly higher than the prior probability of 0.5%, indicating that the observed behavior is a strong indicator of fraud.
- **Conclusion:**
  - By applying Bayes' theorem, the telecom company can more accurately assess the likelihood of fraud based on observed behaviors, allowing them to take appropriate actions, such as flagging the account for further investigation or implementing additional security measures.
  - This approach helps in reducing false positives and improving the efficiency of fraud detection systems.

Probability Math

- **Example: Generative Classifiers**
  - We can **generalize** the medical diagonosis and fraud detection examples to classify feature vectors **x** of arbitrary type as follows:

$$p(y = c|\mathrm{x};\theta) = \frac{p(\mathrm{x}|y = c;\theta)p(y = c;\theta)}{\sum_{c'} p(\mathrm{x}|y = c';\theta)p(y = c;\theta)}$$

  - The term $p(y = c;\ \boldsymbol{\theta})$ is the prior over class labels, and the term $p(\mathbf{x}|y = c;\boldsymbol{\theta})$ is called the **class conditional density** for class c.
  - The overall model is called a **generative classifier**.
  - An alternative approach is to directly fit the class posterior $p(y{=}c|\mathbf{x};\ \boldsymbol{\theta})$; this is known as a **discriminative classifier.**
  - A common example of a discriminative classifier is **logistic regression**.
  - Examples of generative classifiers include **naive Bayes classifier** and **linear discriminant analysis(LDA)**.

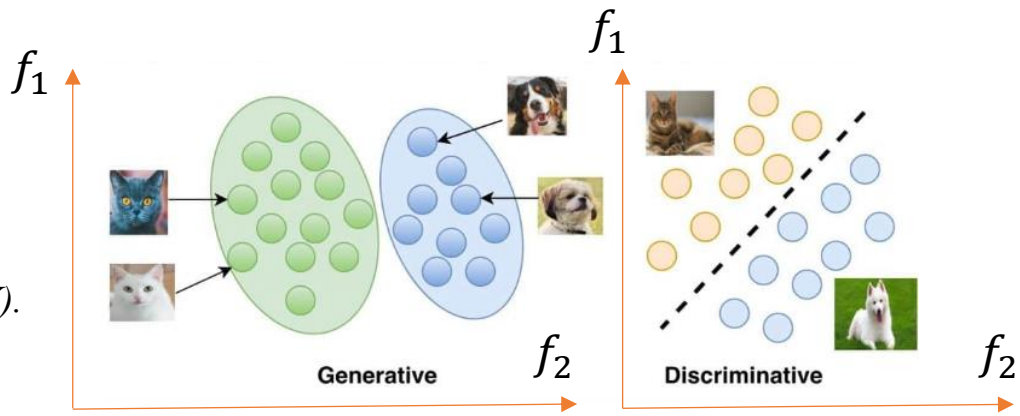
Probability Math

- **Generative Classifier vs. Generative Modeling**
  - **Generative Modeling**
    - **Generative modeling** and **generative classifiers** are related concepts, but they are not exactly the same.
    - Generative modeling is a broader concept that focuses on **learning the underlying probability distribution of data**. Its primary goal is to **generate new data** samples that **resemble** the training data.
    - Generative modeling aims to model the **joint probability distribution** ***P(X,Y)*** of input features X and target variable Y, or just ***P(X)*** if there are no labels.
    - Generative modeling can be used to generate new, **synthetic** data instances that are similar to the training data.
    - Examples include **Generative Adversarial Networks (GANs)**, **Variational Autoencoders (VAEs)**, and other deep learning models capable of generating new data.

Probability Math

- **Generative Classifier vs. Generative Modeling**
  - **Generative Classifiers**
    - Generative classifiers are a specific application of generative modeling principles to classification tasks.
    - They learn a model of the joint probability ***P(X,Y)*** and use Bayes' rule to make predictions.
    - The focus is on using the generative model for classification rather than generating new data.
  - Both are based on modeling the **underlying probability distribution of the data**.
  - **Output**: Generative models produce new instances of data, whereas generative classifiers output class labels based on the learned distributions.

- **Generative Modeling vs. Discriminative Modeling**
  - More formally, given a set of data instances X and a set of labels Y:
    - Generative models capture the joint probability $p(X, Y)$, or just $p(X)$ if there are no labels.
    - Discriminative models capture the conditional probability $p(Y \mid X)$.



Probability Math

- **Independent Events**
  - A and B are called independent if: $P(A \cap B) = P(A)\,P(B)$
  - Therefore: $P(A|B) = P(A)$ and $P(B|A) = P(B)$.





  - **Example:** If we flip a coin in the air and get the outcome as Head, then again if we flip the coin but this time we get the outcome as Tail. From this example, we can see the first event does not affect the occurrence of the next event.

- **Mutually Exclusive vs. Independent Events**:
  - For **independent events** we have two events derived from **two different trials** (two different events like tossing coin and rolling a dice, tossing two coins). Therefore, probability of occurrence of one does not affect the probability of occurrence of other.
  - In case of **mutually exclusive events**, events cannot occur simultaneously. We have also two events (may be more than two) but difference is that the events derived from the **same trial** (in rolling a dice, facing even numbers & odd numbers are mutually exclusive).
  - In case of mutually exclusive events: $A \cap B = \emptyset \rightarrow P(A \cap B) = 0 \rightarrow P(A \cup B) = P(A) + P(B)$.

Probability Math

- **Remember:**
  - Using the mathematical structure of probability, the goal is to perform **automated reasoning**, and in this sense, probability generalizes logical reasoning.
  - When constructing automated reasoning systems, **classical Boolean logic** does not allow us to express certain forms of **plausible reasoning**.
  - For plausible reasoning it is necessary to extend the discrete true and false values of **truth** to continuous plausibilities.
  - In the previous sections, we learned the three concepts that define a **probability space** (the **sample space**, the **events**, and the **probability of an event**).
  - In **machine learning**, we often avoid explicitly referring to the probability space, but instead refer to probabilities on quantities of interest.
- In the next section, we learn how they are related to fourth and fifth concepts called **random variable** and **probability distribution**.

One Random Variable

- **Quantifying uncertainty** requires the idea of a ***random variable***, which is a **function** that maps outcomes of random experiments to a set of properties that we are interested in.

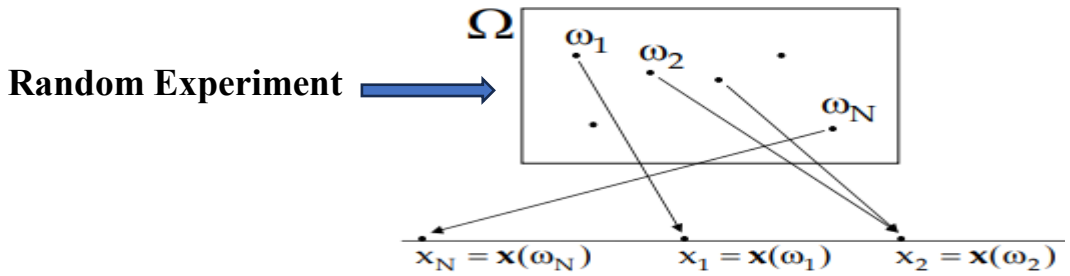
- Associated with the random variable is a **function** that measures the probability that a particular outcome (or set of outcomes) will occur; this is called the ***probability distribution***.

- **Random Variable**

  - The name "random variable" is a great source of misunderstanding, it is not a variable. It is a function.

  - A random variable is a function of sample space points that assigns a number to each of the sample space points.

$$\mathbf{x}(\omega): \Omega \rightarrow \mathbb{R}$$



One Random Variable

- **Example**: in the experiment of throwing three fair coins:
  - $\Omega = \{\underbrace{HHH}_{\omega_1}, \ldots, \underbrace{TTT}_{\omega_8}\}$
  - For example, the number of heads is a random variable.
  - $\mathbf{x}(\omega)$ = number of heads in event ω
  - -

ω	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$\mathbf{x}(\omega)$	3	2	2	1	2	1	1	0



  - Now, in the above example, what is the probability $\mathbf{x}(\omega) \leq 1$?
  - Probability is defined on events:
  - We write: $\{\mathbf{x} \leq x\}$, but this is not $P\{\mathbf{x} \leq 1\} = P\{\omega : \mathbf{x}(\omega) \leq 1\} = P\{HTT, THT, TTH, TTT\} = \frac{4}{8}$

One Random Variable

- **Example**: What is the probability $\mathbf{x}(\omega) = 2$?

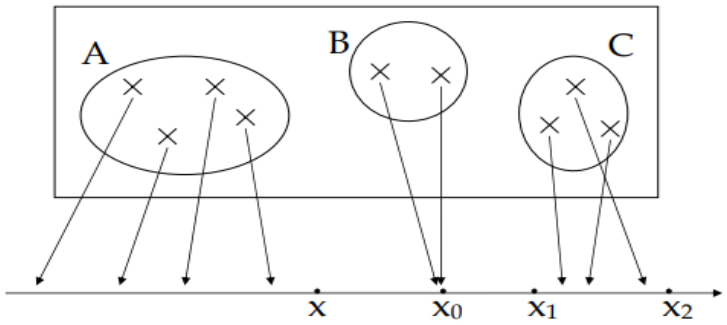
$$P\{\mathbf{x}=2\} = P\{\omega : \mathbf{x}(\omega)=2\} = P\{HHT, HTH, THH\} = \frac{3}{8}$$

- **Random Variable vs. Event**

  - **Definition:**
    - Random Variable: A function that assigns a real number to each outcome in the sample space.
    - Event: A subset of the sample space, representing a collection of possible outcomes.

  - **Purpose:**
    - Random Variable: Used to quantify or measure outcomes of an experiment.
    - Event: Used to describe specific occurrences or conditions within the sample space.

  - **Relationship:**
    - Random variables can be used to define events.
    - For example, $\{\mathbf{x} < x\}$ is an event defined by the random variable.



$A = \{\mathbf{x} \le x\}$
$B = \{\mathbf{x} = x_0\}$
$C = \{x_1 < \mathbf{x} \le x_2\}$

One Random Variable

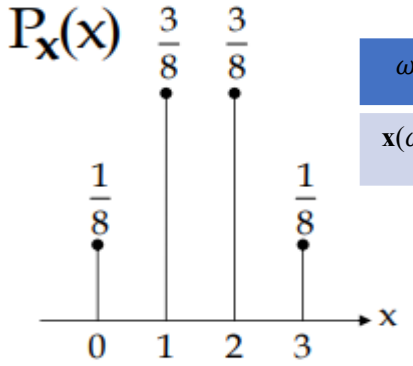
- A random variable is called **discrete** when the values that $\mathbf{x}(\omega)$ can take (the range of the function) can be counted.
- In a discrete state, the simplest way to **determine the probability of events** is to specify the probability of events $\{\mathbf{x} = x_i\}$ for possible x_i (than to specify the probability of $\{\mathbf{x} \leq x\}$ for each x).

▪ Probability Mass Function(pmf)

- If **x** is only possible values of $x_1, x_2, x_3, \ldots$ (Countable) with probabilities of $p_1, p_2, p_3, \ldots$. The probability function of the random variable x is defined as:

$$P_{\mathbf{x}}(x) = Prob(\mathbf{x} = x) = \begin{cases} p_i & \mathbf{x} = x_i \\ 0 & otherwise \end{cases}$$

  - In the previous example:
  - $P\{\mathbf{x} \leq 1.1\} = P\{\mathbf{x} = 0\} + P\{\mathbf{x} = 1\} = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$
  - pmf is defined for discrete random variable.



ω	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$\mathbf{x}(\omega)$	3	2	2	1	2	1	1	0

One Random Variable

- **Cumulative Distribution Function(cdf)**

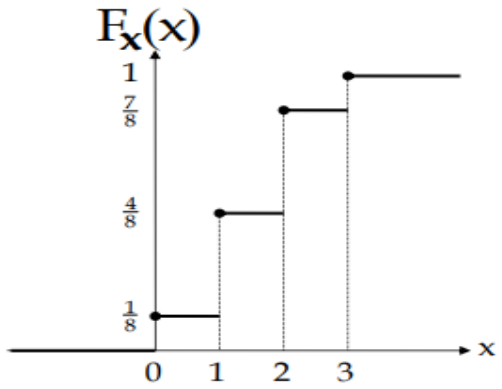
  - When the random variable is **continuous**, i.e., the values it can adopt are uncountable, it is not sufficient to determine the probability $\{\mathbf{x} = x_i\}$. Most of them are zero. Here we can define the **cumulative distribution function**.

$$F_{\mathbf{x}}(x) \triangleq Prob(\mathbf{x} \leq \mathrm{x})$$

  - Whether for a continuous or discrete random variable, if we know the probability of events $\{\mathbf{x} \leq \mathrm{x}\}$ for each x, the probability of all events will be determined. So $F_{\mathbf{x}}(x)$ can fully describe the random variable.

  - In the previous example:

ω	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$\mathbf{x}(\omega)$	3	2	2	1	2	1	1	0



$$F_x(-0.001) = P\{\mathbf{x} \leq -0.001\} = 0$$
$$F_x(0) = P\{\mathbf{x} \leq 0\} = \tfrac{1}{8}$$
$$F_x(0.001) = P\{\mathbf{x} \leq 0.001\} = \tfrac{1}{8}$$
$$F_x(0.999) = P\{\mathbf{x} \leq 0.999\} = \tfrac{1}{8}$$
$$F_x(1) = P\{\mathbf{x} \leq 1\} = \tfrac{4}{8}$$
$$F_x(1.0001) = P\{\mathbf{x} \leq 1.0001\} = \tfrac{4}{8}$$
$$F_x(10) = P\{\mathbf{x} \leq 10\} = 1$$

One Random Variable

- **Probability Density Function(pdf)**
  - For a continuous random variable, another way to determine the probability of events that define the random variable **x** is to have a function that expresses the probability density (similar to what we have defined $\alpha(x)$ directly for the probability of events themselves).
  - With pdf we get more visibility (compared to cdf) relative to the probability of different values.
  - We define pdf as follows:

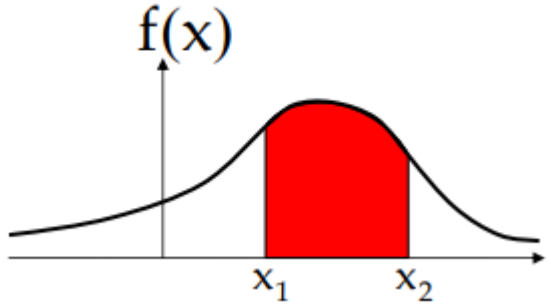
$$f_{\mathbf{x}}(x) \triangleq \frac{dF_{\mathbf{x}}(x)}{dx}$$

  - **Properties** of the probability density function(pdf):

1) $P\{x_1 \le \mathbf{x} \le x_2\} = \int_{x_1}^{x_2} f_{\mathbf{x}}(x)dx$

2) $f(x) \ge 0$

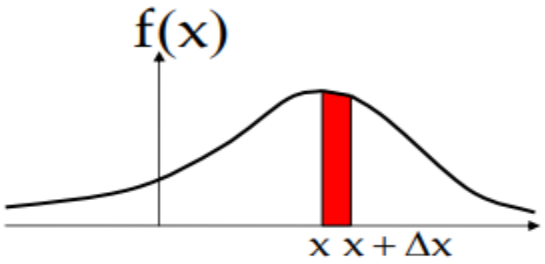
3) $\int_{-\infty}^{+\infty} f(x)dx = 1$ Similar to $\sum_{i=1}^{\infty} P_{\mathbf{x}}(x_i)$



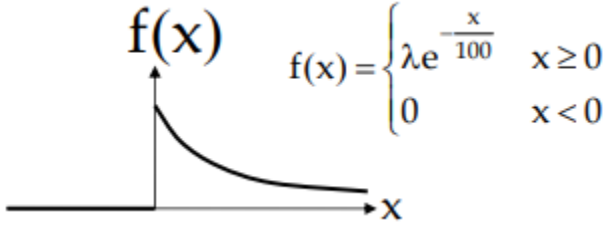
One Random Variable

- **Probability Density Function(pdf)**
  - **The concept of density:**
    - $P\{x \le \mathbf{x} \le x + \Delta x\} = f(x)\Delta x$

$$f(x) = \lim_{\Delta x \to 0} \frac{P\{x \le \mathbf{x} \le x + \Delta x\}}{\Delta x}$$



  - **Example**: The duration of operation of an electrical component (in days) before failure is a continuous random variable with the following density function:



$$f(x) = \begin{cases} \lambda e^{-\frac{x}{100}} & x \ge 0 \\ 0 & x < 0 \end{cases}$$

One Random Variable

- **Probability Density Function(pdf)**
  - How likely is it that the component will run for between 50 and 150 days?

$$\int_{-\infty}^{+\infty} f(x)dx = 1 \Rightarrow \lambda \int_{0}^{+\infty} e^{-\frac{x}{100}} dx = 1 \Rightarrow -\lambda(100)e^{-\frac{x}{100}}\Big|_{0}^{+\infty} = 100\lambda = 1 \Rightarrow \lambda = \frac{1}{100}$$

$$P\{50 < \mathbf{x} < 150\} = \int_{50}^{150} \frac{1}{100} e^{-\frac{x}{100}} dx = -e^{-\frac{x}{100}}\Big|_{50}^{150} = e^{-\frac{1}{2}} - e^{-\frac{3}{2}} = 0.384$$

  - How likely is it that it will work for less than 100 days?

$$P\{\mathbf{x} < 100\} = \int_{0}^{100} \frac{1}{100} e^{-\frac{x}{100}} dx = -e^{-\frac{x}{100}}\Big|_{0}^{100} = 1 - e^{-1} = 0.632$$

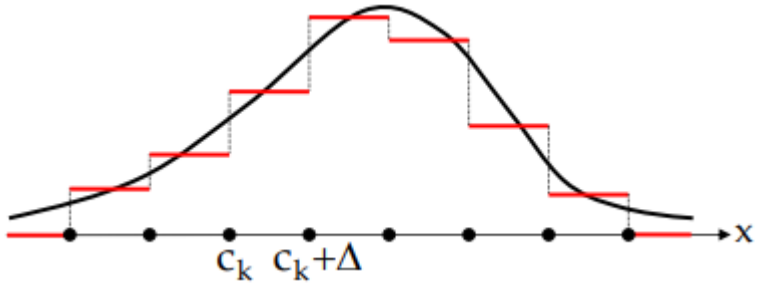
One Random Variable



- **Obtaining pdf experimentally**
  - We repeat the experiment n times and record the value of the random variable **x** obtained from each experiment.
  - We divide the x-axis into Δ-length intervals.
  - If we show the number of x_i that are within the $[c_k, c_k + \Delta)$ with n_k, we have:

$$f(x) \approx f_n(x) = \frac{n_k}{n\Delta} : \; c_k \leq \mathbf{x} < \; c_k + \Delta$$

$$f(c_k)\,\Delta \approx Prob(\; c_k \leq \mathbf{x} < \; c_k + \Delta) \approx \frac{n_k}{n} = \; f_n(x)\Delta$$



One Random Variable

❖ Some common continuous distributions

1. Gaussian (normal) distribution

- The most widely used distribution in statistics and machine learning is the **Gaussian** or **normal distribution**.
- For example, in a homogeneous society, the distribution of height and weight of people is normal.
- According to the **central limit theorem**, the sum of the random variables, even if they are not normal, tends to normal.
- Normal distribution is also mathematically simple to work with.
- Its pdf is given by:

$$\mathbf{x} \sim N(0,1)$$

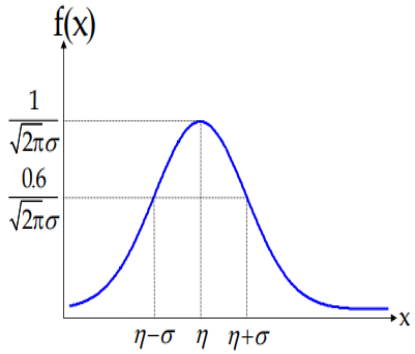
$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = g(x)$$

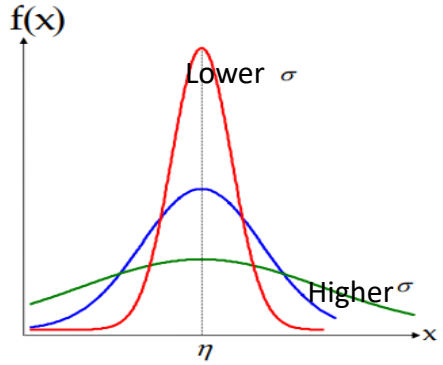
$$F(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{u^2}{2}} du = G(x)$$

$$\mathbf{x} \sim N(\eta,\sigma)$$

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\eta)^2}{2\sigma^2}} = \frac{1}{\sigma} g\left(\frac{x-\eta}{\sigma}\right)$$

$$F_x(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{x} e^{-\frac{(u-\eta)^2}{2\sigma^2}} du = G\left(\frac{x-\eta}{\sigma}\right)$$





One Random Variable

❖ Some common continuous distributions

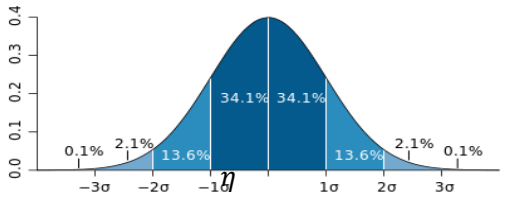
1. Gaussian (normal) distribution

- If $\mathbf{x} \sim N(\eta, \sigma)$, then $a\mathbf{x} + b \sim N(a\eta + b, a\sigma)$, thus $\mathbf{Z} = \frac{\mathbf{x} - \eta}{\sigma} \sim N(0,1)$

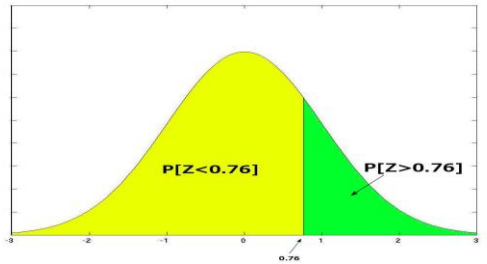
$$P\{x_1 \le \mathbf{x} \le x_2\} = P\{\frac{x_1 - \eta}{\sigma} \le \underbrace{\frac{\mathbf{x} - \eta}{\sigma}}_{z} \le \frac{x_2 - \eta}{\sigma}\} = F_z(\frac{x_2 - \eta}{\sigma}) - F_z(\frac{x_1 - \eta}{\sigma}) = G(\frac{x_2 - \eta}{\sigma}) - G(\frac{x_1 - \eta}{\sigma})$$

$P(-1 \le \boldsymbol{Z} \le 1) = G(1) - G(-1)$

$$P\{\eta - \sigma \le \mathbf{x} \le \eta + \sigma\} \simeq 0.683$$
$$P\{\eta - 2\sigma \le \mathbf{x} \le \eta + 2\sigma\} \simeq 0.954$$
$$P\{\eta - 3\sigma \le \mathbf{x} \le \eta + 3\sigma\} \simeq 0.997$$



For example: P(**Z**<0.76) = G(0.76)



$$P[Z < 0.76]$$

$$0.76 = 0.7 + 0.06$$

- **Cumulative Probability Tables of the Normal Distribution**
  - Once we obtain a z value, we can use the cumulative probability tables of the Normal Distribution to find out the probability that **Z** is less than the z value we have obtained.
  - Python

Z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

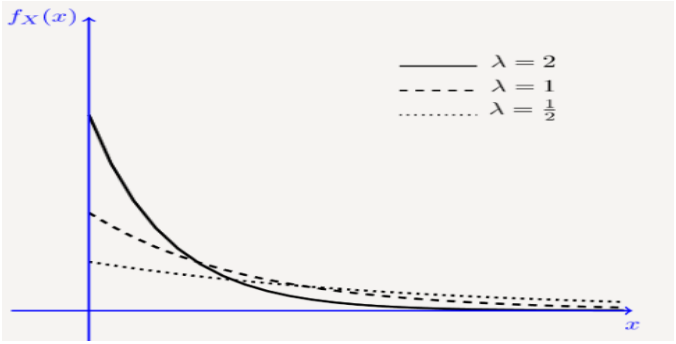
One Random Variable

❖ Some common continuous distributions

2. Exponential Distribution

- A continuous random variable X is said to have an exponential distribution with parameter $\lambda > 0$, shown as X~Exponential(λ) , if its pdf is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$



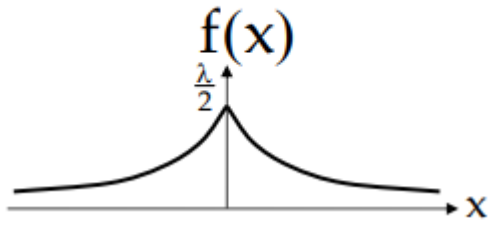
- For example, the duration in minutes of telephone conversation follows the exponential distribution.

One Random Variable

❖ Some common continuous distributions

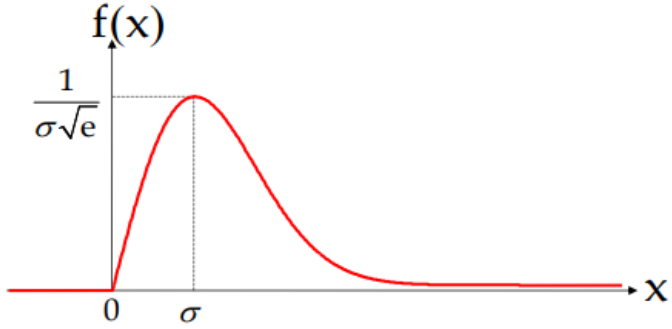
3. Laplace Distribution

$$f(x)=\frac{\lambda}{2}e^{-\lambda|x|}$$



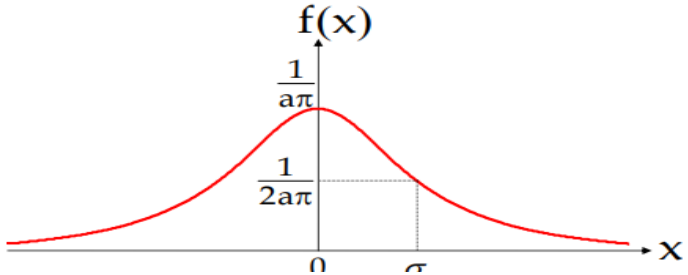
4. Rayleigh Distribution

$$f(x)=\frac{x}{\sigma^2}e^{-\frac{x^2}{2\sigma^2}}u(x)$$

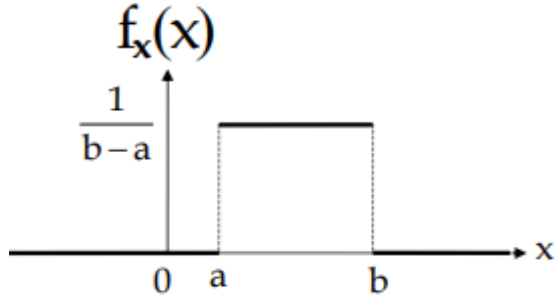


5. Cauchy Distribution

$$f(x)=\frac{\frac{a}{\pi}}{x^2+a^2}$$



6. Uniform Distribution



❖ Some common continuous distributions

6. Uniform Distribution

- For example, the phase of the received signal is often assumed to be uniform between 0 and 2π.

> In cases where we do not have any prior information about distribution, **uniform distribution** is used according to the **principle of insufficient reason.**

- There is a random experiment behind each of these distributions. Since these random experiments **model** a lot of real life phenomenon, these special distributions are used frequently in different applications.

One Random Variable

❖ Some common <u>discrete</u> distributions

1. Binomial Distribution

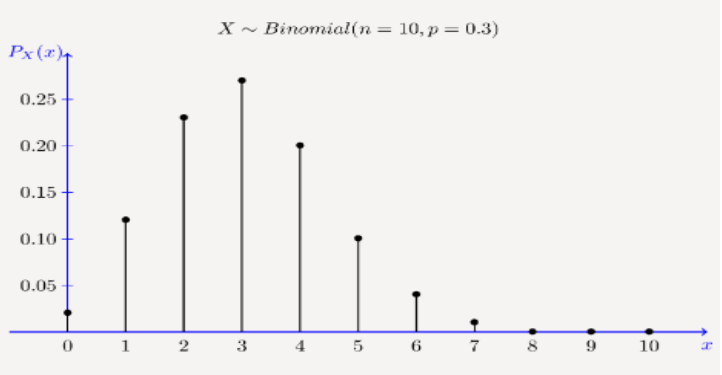
- If in a random experiment the probability of occurrence of event A is equal to P(A) = p and for the experiment resulting from the n repetition of this experiment, we define the random variable X as:

X = the number of occurrence of an event A in n experiment

- X will have a binomial distribution. The **pmf** is given by:

$$X \sim Binomial(n, p)$$

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & \text{for } k = 0, 1, 2, \cdots, n \\ 0 & \text{otherwise} \end{cases}$$



- $\binom{n}{k}$ is the number of ways to choose k items from n.
- The <u>underlying assumptions of binomial distribution</u> are that there is only one outcome for each trial, each trial has the same probability of success, and each trial is mutually exclusive or independent of one another.

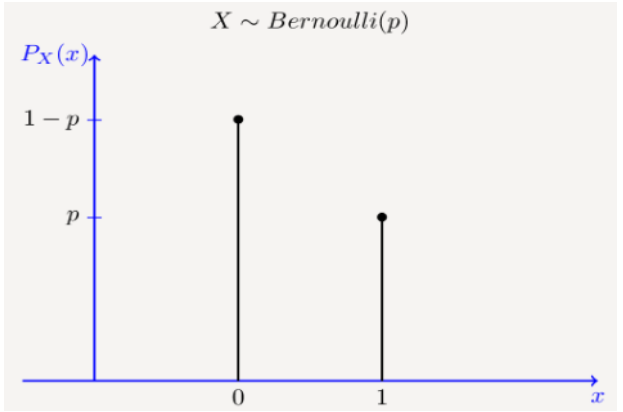
One Random Variable

❖ Some common discrete distributions

2. Bernoulli distribution

- Bernoulli distribution is a special case of a Binomial distribution with n = 1.
- A Bernoulli random variable is a random variable that can only take two possible values, usually 0 and 1. This random variable **models** random experiments that have two possible outcomes, sometimes referred to as "success" and "failure."
- A random variable X is said to be a Bernoulli random variable with parameter p, shown as X~Bernoulli(p), if its **pmf** is given by

$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1-p & \text{for } x = 0 \\ 0 & \text{otherwise} \end{cases}$$



One Random Variable

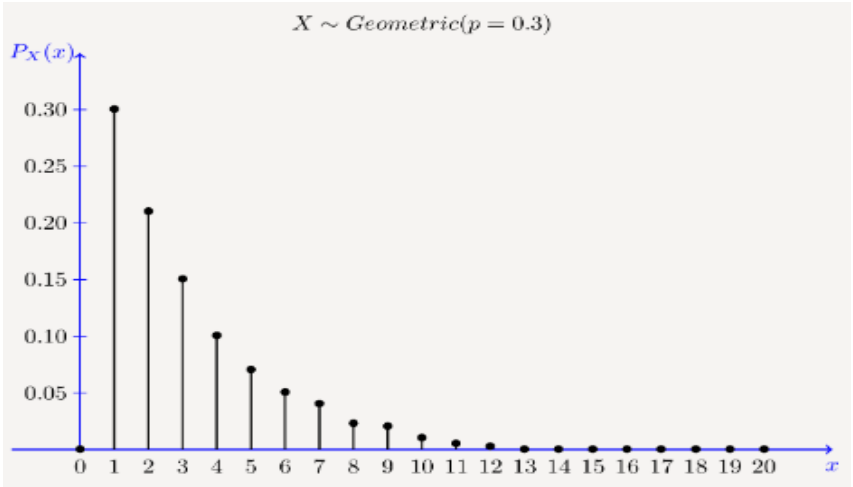
❖ Some common <u>discrete</u> distributions

3. Geometric Distribution

- The random experiment behind the geometric distribution is as follows. Repeating independent Bernoulli trials until observing the first success.

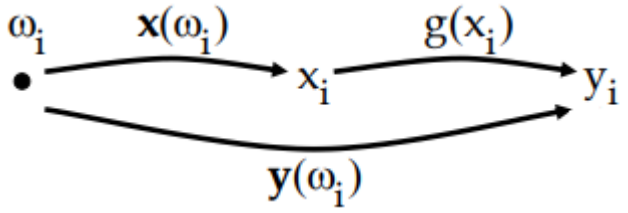
- A random variable X is said to be a geometric random variable with parameter p, shown as X∼Geometric(p), if its **pmf** is given by

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

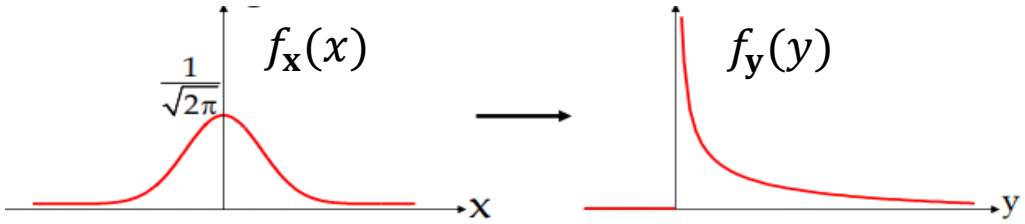


One Random Variable

- **Function of Random Variable**
  - In many cases, assuming that the distribution function of a random variable is known, we would like to obtain the distribution of a **function** of a random variable.



  - New random variable: $\mathbf{y}(\omega) = g(\mathbf{x}(\omega))$
  - We call the random variable **y** a function of the random variable **x**.
  - We can get the distribution **y** by knowing the distribution **x**.
  - For example:



One Random Variable

- **Function of Random Variable**
  - **Example:** If $\mathbf{y} = g(\mathbf{x}) = a\,\mathbf{x} + \mathrm{b}$

$$F_y(y) = P\{y \le y\} = P\{ax + b \le y\} = \begin{cases} P\{x \le \dfrac{y-b}{a}\} = F_x(\dfrac{y-b}{a}) & a > 0 \\ P\{x \ge \dfrac{y-b}{a}\} = 1 - F_x(\dfrac{y-b}{a}) & a < 0 \end{cases}$$

$$f_y(y) = \begin{cases} \dfrac{1}{a} f_x(\dfrac{y-b}{a}) & a > 0 \\ -\dfrac{1}{a} f_x(\dfrac{y-b}{a}) & a < 0 \end{cases} = \frac{1}{|a|} f_x(\frac{y-b}{a})$$

One Random Variable

❖ Mean and Variance of Random Variable

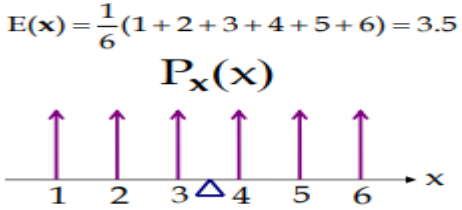
- What completely determined the random variable was its **pdf** or **cdf**.
- **Mean** and **variance** are two important parameters describing a random variable.
- **Mean or Expectation(**denoted by $\eta \; or \; \mu$)

$$\mathrm{E}(\mathbf{x}) \triangleq \int_{-\infty}^{+\infty} x \, f_{\mathbf{x}}(x) \, dx$$

  - For discrete case:

$$\mathrm{E}(\mathbf{x}) \triangleq \sum_i x_i P_{\mathbf{x}}(x_i)$$

  - **Example:** rolling a dice



One Random Variable

❖ Mean and Variance of Random Variable



○ Empirical interpretation of mean:

- If **x** is a random variable with possible values x_i, and we perform the experiment n times, and each x_i is observed n_i times:

$$\text{Arithmetic Mean: } \bar{\mathbf{x}} = \frac{n_1x_1 + n_2x_2 + \ldots + n_kx_k}{n} = \sum_{i=1}^{k} x_i \frac{n_i}{n} \approx \sum_{i=1}^{k} x_i p_i = E(\mathrm{x})$$

❑ **Theorem:** if **y** = g(**x**):

$$E(\mathbf{y}) = \int_{-\infty}^{+\infty} y f_{\mathbf{y}}(y)dy = \int_{-\infty}^{+\infty} g(x) f_{\mathbf{x}}(x)dx$$

- For discrete case:

$$E(\mathbf{y}) = \sum_{i} g(x_i) P_{\mathbf{x}}(x_i)$$

One Random Variable

❖ Mean and Variance of Random Variable

○ Variance(denoted by σ^2)

- The average showed how much **x** values were around.
- We need another quantity that shows us how much **x** values are scattered around the mean value.
- The **variance** represents the extent of the probability dispersion around the mean value.

$$\text{var}(\mathbf{x}) \triangleq \text{E}[\,(\mathbf{x} - E(\mathbf{x}))^2] = \int (x - \eta)^2 f_{\mathbf{x}}(x)dx = \text{E}(\mathbf{x}^2) - \eta^2$$

- For discrete case:

$$\text{var}(\mathbf{x}) \triangleq \sum_i (x_i - \eta)^2 P_{\mathbf{x}}(x_i)$$

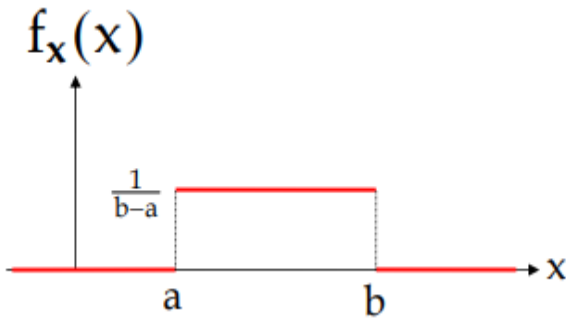
- The **standard deviation** is defined as

$$\text{std}(\mathbf{x}) \triangleq \sqrt{\text{var}(\mathbf{x})}$$

One Random Variable

❖ Mean and variance of some specific distributions

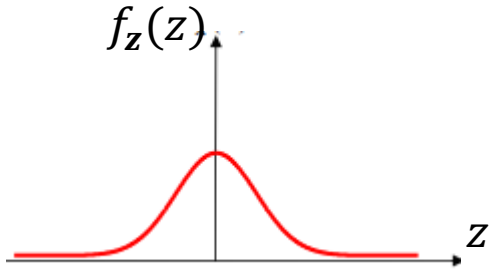
▪ Uniform Distribution



$$E(x)=\int_{-\infty}^{+\infty} x f_x(x)dx=\int_a^b x\frac{1}{b-a}dx=\frac{a+b}{2}$$

$$var(x)=\int_{-\infty}^{+\infty}(x-\eta_x)^2 f_x(x)dx=\int_a^b(x-\frac{a+b}{2})^2\frac{1}{b-a}dx=\frac{(b-a)^2}{12}$$

▪ Normal Distribution



$$z\sim N(0,1)\rightarrow f_z(z)=\frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}$$

$$\eta_z=E(z)=0$$

$$var(z)=E((z-0)^2)=E(z^2)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}z^2e^{-\frac{z^2}{2}}dz=1$$

One Random Variable

❖ Mean and variance of some specific distributions

▪ Bernoulli Distribution

$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1-p & \text{for } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(\mathbf{x}) = 1 \times P(x = 1) + 0 \times P(x = 0) = p$$

$$E(\mathbf{x^2}) = 1^2 \times P(x = 1) + 0^2 \times P(x = 0) = p$$

$$var(\mathbf{x}) = E(\mathbf{x^2}) - E^2(\mathbf{x}) = \mathrm{p} - p^2$$

▪ Binomial Distribution

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & \text{for } k = 0, 1, 2, \cdots, n \\ 0 & \text{otherwise} \end{cases}$$

$$\eta = \mathrm{E}(\mathbf{x}) = \mathrm{np}$$

$$\sigma^2 = \mathrm{var}(\mathbf{x}) = \mathrm{npq}$$

Two Random Variables

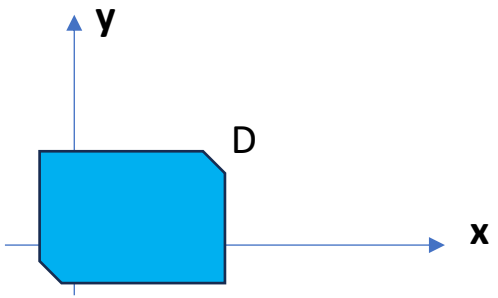
- Random variable **x:**

$$x_1 = \mathbf{x}(\omega_1), x_2 = \mathbf{x}(\omega_2), \ldots, x_N = \mathbf{x}(\omega_N)$$

- Random variable **y:**

$$y_1 = \mathbf{y}(\omega_1), y_2 = \mathbf{y}(\omega_2), \ldots, y_N = \mathbf{y}(\omega_N)$$

- Obtaining probability of events such as $\{\mathbf{x} = x_1, \mathbf{y} = y_1\}$ *or* $\{\mathbf{x} \leq x_0, \mathbf{y} \leq y_0\}$ and in general, $\{(\mathbf{x}, \mathbf{y})\} \in D$ means $\{\omega: (\mathbf{x}(\omega), \mathbf{y}(\omega))\} \in D$.



- In general, just by having distribution of **x** and distribution of **y**, the probability of these (two-dimensional) events is not determined (unless **x** and **y** are independent), and we need new tools to determine the probability of these events.

Two Random Variables

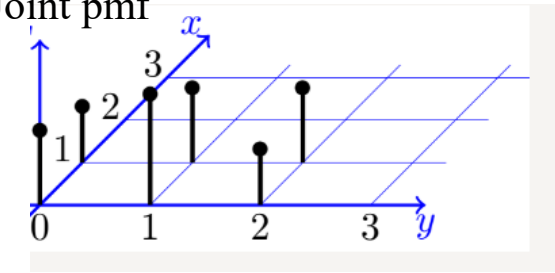
- **Joint Probability Function or Joint pmf**
  - If **x** and **y** are discrete random variables, then:

$$P_{xy}(x,y) = P\{\mathbf{x}=x, \mathbf{y}=y\}$$

$$P_{xy}(x,y) = \begin{cases} p_{ij} & x = x_i, y = y_i \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_i \sum_j P_{xy}(x_i, y_j) = \sum_i \sum_j p_{ij} = 1$$

Joint pmf



  - We know that $\{\mathbf{x}=x_i\} = \bigcup_j \{\mathbf{x}=x_i, \mathbf{y}=y_j\}$,thus

$$P\{\mathbf{x}=x_i\} = \sum_j P\{\mathbf{x}=x_i, \mathbf{y}=y_j\}$$

Two Random Variables

- **Joint Probability Function or Joint pmf**
  - **Marginal pmf:**
    - Using law of total probability

$$\begin{cases} P_x(x_i) = \sum_j P_{xy}(x_i, y_j) & \text{Marginal pmf of } \mathbf{x} \\ P_y(y_j) = \sum_i P_{xy}(x_i, y_j) & \text{Marginal pmf of } \mathbf{y} \end{cases}$$

  - For each subset D from $\mathbb{R}^2$:

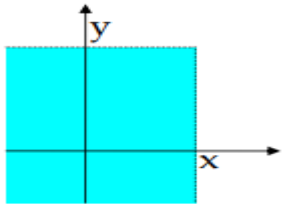
$$P\{(\mathbf{x}, \mathbf{y}) \in D\} = \sum_{(x_i, y_j) \in D} P(x_i, y_j)$$

  - For continuous random variable $P\{\mathbf{x} = x_1, \mathbf{y} = y_j\} = 0$, therefore pmf can not determine probability of events. Thus, we need to define new tools.

Two Random Variables

- **Joint cdf**
  - **Definition:**

$$F_{\mathbf{x},\mathbf{y}}(x,y) \triangleq Prob(\mathbf{x} \le \mathrm{x}, \mathbf{y} \le \mathrm{y})$$



  - For discrete case:

$$F_{\mathbf{xy}}(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} P_{\mathbf{xy}}(x_i, y_j)$$

- **Joint pdf**

$$f_{\mathbf{x},\mathbf{y}}(x,y) \triangleq \frac{\partial^2 F_{\mathbf{x},\mathbf{y}}(x,y)}{\partial x \partial y}$$

  - From definition:

$$f_{\mathbf{xy}}(x,y)dxdy = P\{x < \mathbf{x} \le x + dx, y < \mathbf{y} \le y + dy\}$$

  - Therefore,

$$P\{(\mathbf{x},\mathbf{y}) \in D\} = \iint_D f_{\mathbf{xy}}(x,y)dxdy$$

i.e. the volume below the function $f_{\mathbf{x},\mathbf{y}}(x,y)$ in each area, indicates the probability of that area

Two Random Variables

- **Joint pdf**
  - Property:

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)\,dx\,dy = F(+\infty,+\infty) = 1$$

  - **Marginal pdf:**

$$f_x(x) = \int_{-\infty}^{+\infty} f_{xy}(x,y)\,dy$$ Marginal pdf of **x**

$$f_y(y) = \int_{-\infty}^{+\infty} f_{xy}(x,y)\,dx$$ Marginal pdf of **y**

- **Independent Random Variable**
  - Two random variables **x** and **y** are called **independent**, if for each x and y, the events $\{\mathbf{x} \le x\}$ and $\{\mathbf{y} \le y\}$ are independent,

$$P\{\mathbf{x} \le x, \mathbf{y} \le y\} = P\{\mathbf{x} \le x\}P\{\mathbf{y} \le y\}$$

  - In other symbols

$$F_{xy}(x,y) = F_x(x)F_y(y) \xrightarrow{\frac{\partial^2}{\partial x \partial y}} f_{xy}(x,y) = f_x(x)f_y(y)$$

Two Random Variables

- **Independent Random Variable**
  - For discrete case:

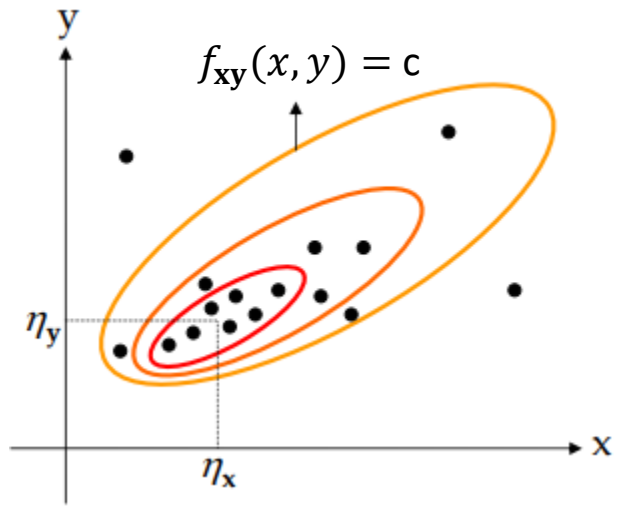
$$\underbrace{P\{\mathbf{x}=x_i,\mathbf{y}=y_j\}}_{\downarrow\ P_{xy}(x_i,y_j)} = \underbrace{P\{\mathbf{x}=x_i\}}_{\downarrow\ P_x(x_i)}\underbrace{P\{\mathbf{y}=y_j\}}_{\downarrow\ P_y(y_j)}$$

- **Expectation, Covariance and Correlation**
  - **Expectation:**
    - **Theorem:**

$$E(g(\mathbf{x},\mathbf{y})) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(x,y) f_{xy}(x,y)\,dx\,dy$$

$$E(\mathbf{x}) = \eta_x = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} x f_{xy}(x,y)\,dx\,dy = \int_{-\infty}^{+\infty} x f_x(x)\,dx$$

$$\sigma_x^2 = E((\mathbf{x}-\eta_x)^2) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} (x-\eta_x)^2 f_{xy}(x,y)\,dx\,dy = \int_{-\infty}^{+\infty} (x-\eta_x)^2 f_x(x)\,dx$$



  - And for η_y *and* σ_y^2: …

Two Random Variables

- **Expectation, Covariance and Correlation**
  - Definition**: Covariance**

$$\operatorname{cov}(\mathbf{x},\mathbf{y}) = \mu_{xy} = \mathrm{E}\left[(\mathbf{x}-\eta_x)(\mathbf{y}-\eta_y)\right] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(x-\eta_x)(y-\eta_y)f_{xy}(x,y)\,dx\,dy$$

    - **Properties:**

1. -

$$\operatorname{cov}(\mathbf{x},\mathbf{y}) = \mathrm{E}\left[(\mathbf{x}-\eta_x)(\mathbf{y}-\eta_y)\right] = \mathrm{E}(\mathbf{xy}) - \eta_x\mathrm{E}(\mathbf{y}) - \eta_y\mathrm{E}(\mathbf{x}) + \eta_x\eta_y$$
$$\Rightarrow \operatorname{cov}(\mathbf{x},\mathbf{y}) = \mathrm{E}(\mathbf{xy}) - \mathrm{E}(\mathbf{x})\mathrm{E}(\mathbf{y})$$

2. -

$$\operatorname{cov}(\mathbf{x},\mathbf{x}) = \mathrm{E}((\mathbf{x}-\eta_x)^2)$$
$$\Rightarrow \operatorname{cov}(\mathbf{x},\mathbf{x}) = \operatorname{var}(\mathbf{x})$$

  - Definition**: Correlation Coefficient(Pearson correlation coefficient)**

$$\rho_{xy} = \frac{\operatorname{cov}(\mathbf{x},\mathbf{y})}{\sqrt{\operatorname{var}(\mathbf{x})\operatorname{var}(\mathbf{y})}} = \frac{\mu_{xy}}{\sigma_x\sigma_y}$$

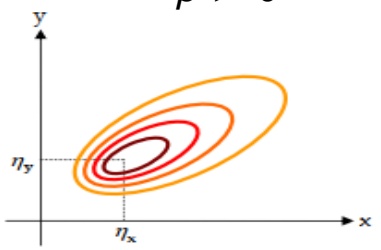
Two Random Variables

- **Expectation, Covariance and Correlation**
  - Definition: **Correlation Coefficient(Pearson correlation coefficient)**
    - It can be shown that:

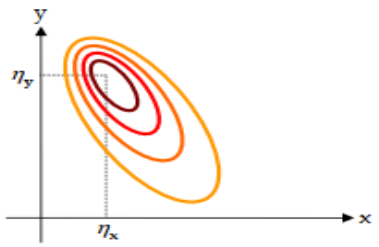
$$\rho_{xy} = \mathrm{cov}\left(\frac{x-\eta_x}{\sigma_x}, \frac{y-\eta_y}{\sigma_y}\right)$$

    - and $-1 \leq \rho_{xy} \leq 1$.
    - If **x** goes above its average, **y** will usually be above its average. In this case: $\rho_{xy} > 0$;
    - On the contrary, if **x** goes above its average, **y** will be lower than its average. In this case: $\rho_{xy} < 0$;
    - If **x** and **y** don't affect each other. In this case: $\rho_{xy} = 0$.

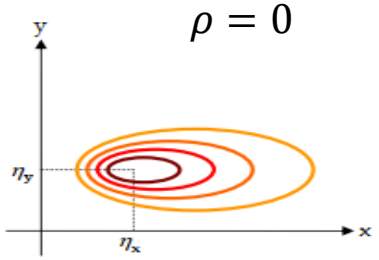
$\rho > 0$



$\rho < 0$



$\rho = 0$



Example: ...

Two Random Variables

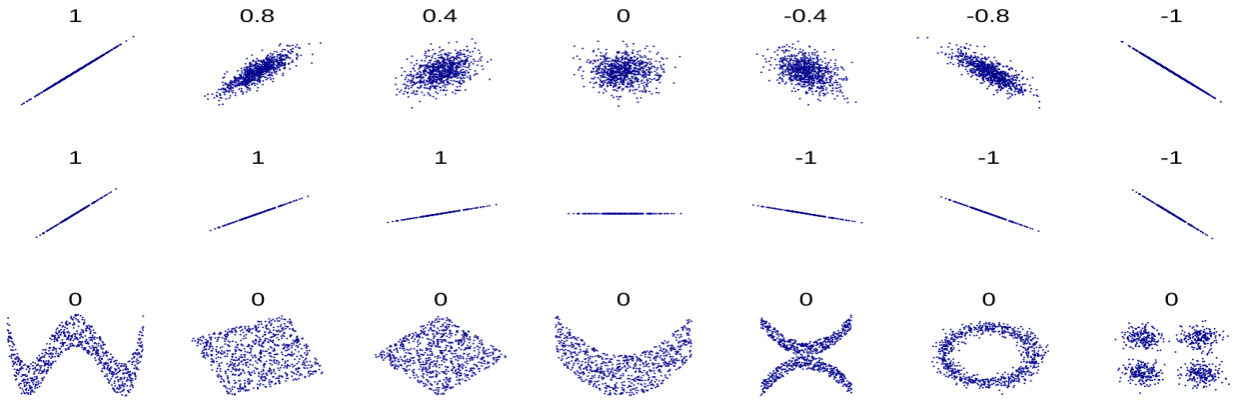
- **Expectation, Covariance and Correlation**
  - **Correlation Coefficient(Pearson correlation coefficient)**
    - The **Pearson correlation coefficient** (**PCC**) is a correlation coefficient that measures **linear** correlation between two sets of data(two random variables).
    - As with covariance itself, the **measure** can only reflect a **linear correlation** of variables, and ignores many other types of **relationships** or **correlations**.
    - **For population**
      - Pearson's correlation coefficient when applied to a **population,** may be referred to as the **population correlation coefficient** or the **population Pearson correlation coefficient**.
    - **For Sample**



      - Pearson's correlation coefficient, when applied to **sample** and may be referred to as the **sample correlation coefficient** or the **sample Pearson correlation coefficient**.
      - We can obtain it by substituting **estimates** of the covariances and variances based on a sample into the formula.

Two Random Variables

- **Expectation, Covariance and Correlation**
  - **Correlation Coefficient(Pearson correlation coefficient)**
    - Several sets of (*x, y*) points, with the correlation coefficient of **x** and **y** for each set.



    - The correlation reflects the **strength** and **direction** of a linear relationship (top row), but not the slope of that relationship (middle), nor many aspects of nonlinear relationships (bottom).
    - The figure in the center has a slope of 0 but in that case the correlation coefficient is undefined because the variance of **y** is zero.

Two Random Variables

- **Expectation, Covariance and Correlation**
  - **Correlation Coefficient(Pearson correlation coefficient)**
    - **Definition**: Random variables **x** and **y** are called **uncorrelated** if $r_{xy} = 0$,

$$\mathrm{cov}(\mathbf{x},\mathbf{y}) = 0$$

$$R_{xy} = E(\mathbf{xy}) = E(\mathbf{x})E(\mathbf{y})$$

    - **Definition**: Random variables **x** and **y** are called **orthogonal** when:

$$R_{xy} = E(\mathbf{xy}) = 0$$

❑ **Theorem**: If **x** and **y** are **independent**, they will be uncorrelated (but the opposite is not necessarily true), because:

$$\begin{aligned} E(\mathbf{xy}) &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xyf_{xy}(x,y)dxdy \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xyf_x(x)f_y(y)dxdy \\ &= \left(\int_{-\infty}^{+\infty} xf_x(x)dx\right)\left(\int_{-\infty}^{+\infty} yf_y(y)dy\right) \\ &= E(\mathbf{x})E(\mathbf{y}) \quad \Rightarrow \text{ } \mathbf{x} \text{ and } \mathbf{y} \text{ are called uncorrelated} \end{aligned}$$

Conditional Distributions

- We recall that **conditional probability** of an **event** is given by:

$$P(A \mid M) = \frac{P(A \cap M)}{P(M)}$$

- **Conditional Probability Function (Conditional pmf)**

  - If **x** is a discrete random variable with values x_i , $i = 0, 1, 2, \ldots$, the **conditional probability function** **x** under the condition of the event M is, by definition, equal to:

$$P_{\mathbf{x}}(x_i \mid M) = P\{\mathbf{x} = x_i \mid M\} = \frac{P\{\mathbf{x} = x_i, M\}}{P(M)} = \frac{P\{\omega : \mathbf{x}(\omega) = x_i, \omega \in M\}}{P(M)}$$

- **Conditional Cumulative Distribution Function (Conditional cdf)**

  - By definition, the **conditional cumulative distribution function** **x** under the condition of the event M is equal to:

$$F_{\mathbf{x}}(x \mid M) = P\{\mathbf{x} \le x \mid M\} = \frac{P\{\mathbf{x} \le x,\ M\}}{P(M)}$$

Conditional Distributions

- **Conditional Probability Density Function (Conditional pdf)**
  - By definition, the **conditional density function** **x** is equal to:

$$f_x(x|M)=\frac{dF_x(x|M)}{dx}=\lim_{\Delta x\to 0}\frac{P(x\le \mathbf{x}\le x+\Delta x|M)}{\Delta x}$$

  - This function also has all the properties of unconditional pdf, including:

$$1)\ f_x(x|M)dx=P\{x\le \mathbf{x}\le x+dx|M\}$$

$$2)\ \int_{-\infty}^{+\infty} f_x(x|M)dx=F(+\infty|M)-F(-\infty|M)=1$$

  - If **x** and **y** are **discrete random variable**, then

$$P_y(y_j|x_1<\mathbf{x}\le x_2)=\frac{P\{\mathbf{y}=y_j,x_1<\mathbf{x}\le x_2\}}{P\{x_1<\mathbf{x}\le x_2\}}=\frac{\sum_{x_1<x_i\le x_2}P_{xy}(x_i,y_j)}{\sum_{x_1<x_i\le x_2}P_x(x_i)}$$

and

$$P_y(y_j|\mathbf{x}=x_i)=P_y(y_j|x_i)=\frac{P\{\mathbf{y}=y_j,\mathbf{x}=x_i\}}{P\{\mathbf{x}=x_i\}}=\frac{P_{xy}(x_i,y_j)}{P_x(x_i)}$$

Conditional Distributions

- **Conditional Probability Density Function (Conditional pdf)**
  - And for pdf:

$$f_y(y \mid x_1 < \mathbf{x} \le x_2) = \frac{\int_{x_1}^{x_2} f_{xy}(x,y)dx}{\int_{x_1}^{x_2} f_x(x)dx} \quad \text{and} \quad f_y(y \mid x) = \frac{f_{xy}(x,y)}{f_x(x)}$$

  - If **x** and **y** are independent, then $f_{xy}(x,y) = f_x(x)f_y(y)$, therefore

$$f_y(y \mid x) = f_y(y) \quad \text{and} \quad f_x(x \mid y) = f_x(x)$$

  - We know that

$$f_y(y) = \int_{-\infty}^{+\infty} f_{xy}(x,y)dx \qquad f_{xy}(x,y) = f_y(y \mid x)f_x(x)$$

$$f_y(y) = \int_{-\infty}^{+\infty} f_y(y \mid x)f_x(x)dx$$

Density version of total probability

Conditional Distributions

- **Conditional Probability Density Function (Conditional pdf)**
  - Now, by switching the roles of **x** and **y**, we have

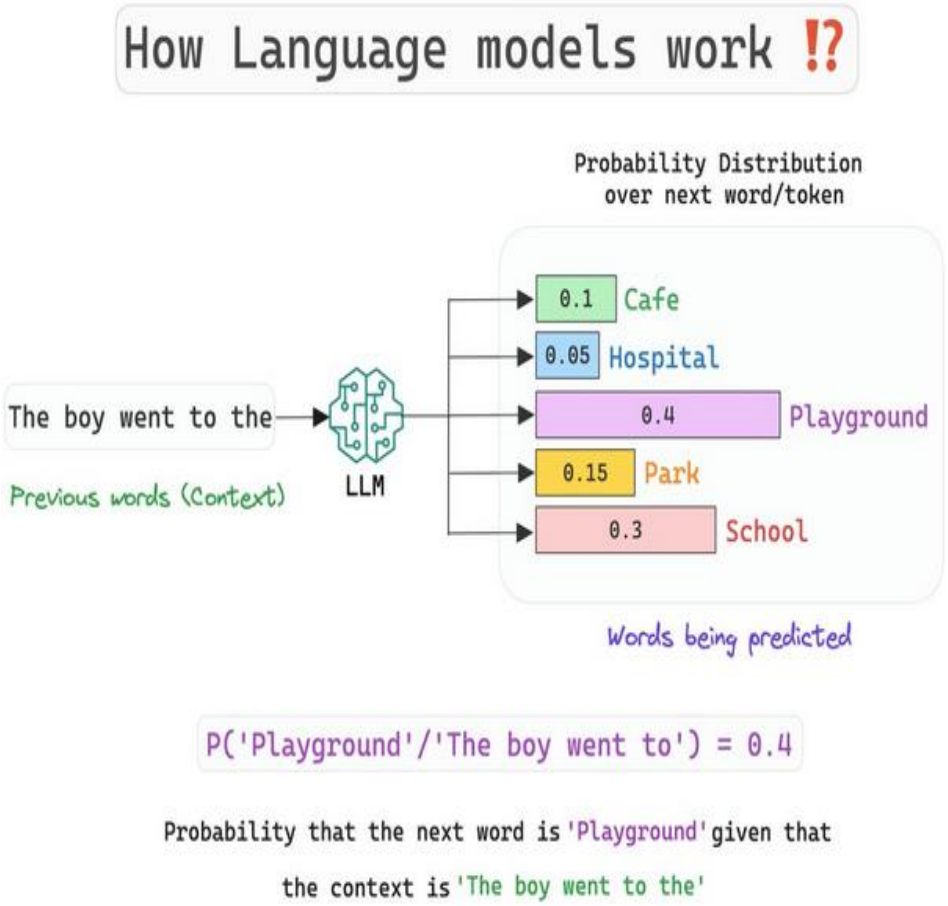
$$f_x(x|y) = \frac{f_{xy}(x,y)}{f_y(y)}$$

$$\Rightarrow f_x(x|y) = \frac{f_y(y|x)f_x(x)}{f_y(y)} = \frac{f_y(y|x)f_x(x)}{\int_{-\infty}^{+\infty} f_y(y|x)f_x(x)dx}$$

Density version of Bayes' theorem

Conditional Distributions

- **Conditional Probability and Its Connection to LLMs**



This is achieved through training on large corpora of text, where the model learns to approximate these conditional probabilities using **neural network architectures**. The training process involves adjusting the model parameters to minimize the difference between the predicted probabilities and the actual occurrences in the training data.

In **transformer**-based LLMs, the conditional probabilities are computed using **attention mechanisms**, which can be seen as a way to dynamically weight the importance of different parts of the input sequence.

while transformers don't directly compute conditional probabilities in the traditional statistical sense, they model these distributions implicitly through their architecture and training process. The softmax output at each step represents the conditional probability distribution of the next token given the previous tokens. This allows LLMs to generate coherent text by repeatedly sampling from these learned conditional distributions.

Conditional Distributions

- **Conditional Expectation**
  - We know that

$$E(\mathbf{x}) = \int_{-\infty}^{+\infty} x f_x(x) dx$$

$$E(g(\mathbf{x})) = \int_{-\infty}^{+\infty} g(x) f_x(x) dx$$

  - In the conditional state, the conditional density function is replaced, i.e.:

$$E(\mathbf{x} \mid M) = \int_{-\infty}^{+\infty} x f_x(x \mid M) dx$$

$$E(g(\mathbf{x}) \mid M) = \int_{-\infty}^{+\infty} g(x) f_x(x \mid M) dx$$

  - **Properties:**

1. $E(E(\mathbf{y} \mid \mathbf{x})) = E(\mathbf{y})$

Proof:

$$E(\underbrace{E(\mathbf{y} \mid \mathbf{x})}_{\Phi(x)}) = \int_{-\infty}^{+\infty} \underbrace{E(\mathbf{y} \mid x)}_{\Phi(x)} f_x(x) dx = \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} y f_y(y \mid x) dy \right) f_x(x) dx$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f_{xy}(x, y) dx dy = E(\mathbf{y})$$

Conditional Distributions

- **Conditional Expectation**
  - **Properties:**

2. If **x** and **y** are independent, then $E(\mathbf{y}\,|\,\mathbf{x}) = E(\mathbf{y})$

Proof:

$$\forall x : E(\mathbf{y}\,|\,x) = \int_{-\infty}^{+\infty} y\, f_{\mathbf{y}}(y\,|\,x)\,dy = \int_{-\infty}^{+\infty} y\, f_{\mathbf{y}}(y)\,dy = \eta_{\mathbf{y}}$$

$$\Rightarrow E(\mathbf{y}\,|\,\mathbf{x}) = \eta_{\mathbf{y}}$$

Sequences of Random Variables

- In the previous sections, we saw how to **generalize** the concepts of one random variable into two random variables.
- Similarly, when we have several random variables, we can bring up past concepts (joint pmf, joint cdf, joint pdf, etc.).

- First, we define the **random vector $\underline{\mathbf{x}}$** as follows, whose components $\mathbf{x}_i$ are random variables.

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

- **Joint pmf**

$$P_{\underline{\mathbf{x}}}(\underline{x}) = P_{\mathbf{x}_1\mathbf{x}_2\cdots\mathbf{x}_n}(x_1, x_2, \ldots, x_n) = P\{\mathbf{x}_1 = x_1, \mathbf{x}_2 = x_2, \ldots, \mathbf{x}_n = x_n\}$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Sequences of Random Variables

- **Joint cdf**

$$F_{\underline{x}}(\underline{x}) = F_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) = P\{x_1 \le x_1, x_2 \le x_2, \ldots, x_n \le x_n\}$$

- **Joint pdf**

$$f_{\underline{x}}(\underline{x}) = f_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) = \frac{\partial^n}{\partial x_1 \partial x_2 \cdots \partial x_n} F_{\underline{x}}(\underline{x}) = \frac{\partial^n}{\partial x_1 \partial x_2 \cdots \partial x_n} F_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n)$$

  - In this way, the following relationships will be established:

$$f_{\underline{x}}(\underline{x}) dx_1 dx_2 \cdots dx_n = P\{x_1 \le x_1 \le x_1 + dx_1, x_2 \le x_2 \le x_2 + dx_2, \ldots, x_n \le x_n \le x_n + dx_n\}$$

$$F_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_n} f_{x_1 x_2 \cdots x_n}(u_1, u_2, \ldots, u_n) du_n \cdots du_2 du_1$$

  - The value of the pdf function is always positive. Its n-integral from $\infty-$ to $+\infty$ is 1. We also have

$$\forall D \subset \mathbb{R}^n : P\{(x_1, x_2, \ldots, x_n) \in D\} = \int_D \cdots \int f_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) dx_1 \cdots dx_n = \int_D f_{\underline{x}}(\underline{x}) d\underline{x}$$

  - If we take an integral from $\infty-$ to $+\infty$ on some of the arguments, we get the joint pdf of the other random variables. For example

$$f_{x_2 x_3}(x_2, x_3) = \int_{-\infty}^{+\infty} f_{x_1 x_2 x_3}(x_1, x_2, x_3) dx_1$$

Sequences of Random Variables

- **Independence**
  - The random variables $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are called (mutually) **independent** if the events $\{\mathbf{x}_1 \leq x_1\}, \ldots, \{\boldsymbol{x}_n \leq x_n\}$ are independent. From this it follows that

$$F_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) = F_{x_1}(x_1) F_{x_2}(x_2) \cdots F_{x_n}(x_n) = \prod_{i=1}^{n} F_{x_i}(x_i)$$

$$f_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) = f_{x_1}(x_1) f_{x_2}(x_2) \cdots f_{x_n}(x_n) = \prod_{i=1}^{n} f_{x_i}(x_i)$$

- **Expectation, Covariance and Correlation**
  - If $\boldsymbol{z} = g(\underline{\mathbf{x}})$:

$$E(\mathbf{z}) = \int_{-\infty}^{+\infty} z f_{\mathbf{z}}(z) dz$$

  - ❑ **Theorem:**

$$E(\mathbf{z}) = E[g(\underline{\mathbf{x}})] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(\underline{x}) f_{\underline{\mathbf{x}}}(\underline{x}) d\underline{x}$$

Sequences of Random Variables

- **Expectation, Covariance and Correlation**
  - To understand how each $\mathbf{x}_i$ is distributed, σ_i and η_i help us to get a general idea of how it is distributed.

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \rightarrow \eta_{\underline{\mathbf{x}}} = \mathrm{E}(\underline{\mathbf{x}}) = \begin{bmatrix} \mathrm{E}(\mathbf{x}_1) \\ \mathrm{E}(\mathbf{x}_2) \\ \vdots \\ \mathrm{E}(\mathbf{x}_n) \end{bmatrix} = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \end{bmatrix}$$

  - And we have seen that the covariance of two random variables showed how they depend on each other, now we are facing the covariance of these n random variables.

$$\sigma_{ij} \triangleq \mu_{\mathbf{x}_i\mathbf{x}_j} = \mathrm{cov}(\mathbf{x}_i, \mathbf{x}_j) = \mathrm{E}\left[(\mathbf{x}_i - \eta_i)(\mathbf{x}_j - \eta_j)\right] : i, j : 1, 2, \ldots, n$$

$$\sigma_{ii} = \mathrm{cov}(\mathbf{x}_i, \mathbf{x}_i) = \mathrm{var}(\mathbf{x}_i) = \sigma_i^2$$

  - **Definition.** covariance matrix of the random vector **x** is equal to

$$C_{\underline{\mathbf{x}}} = \mathrm{E}\left[(\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}}) \cdot (\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}})^{\mathrm{T}}\right] = \mathrm{E}\left(\begin{bmatrix} \mathbf{x}_1 - \eta_1 \\ \mathbf{x}_2 - \eta_2 \\ \vdots \\ \mathbf{x}_n - \eta_n \end{bmatrix} \cdot \begin{bmatrix} \mathbf{x}_1 - \eta_1 & \mathbf{x}_2 - \eta_2 & \cdots & \mathbf{x}_n - \eta_n \end{bmatrix}\right) = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{bmatrix}$$

Sequences of Random Variables

- **Expectation, Covariance and Correlation**
  - Sometimes it is also shown with $C_{\underline{\mathbf{xx}}}$ and it is called **Auto Covariance.**
  - **Properties of Covariance Matrix**
    1. It is **symmetric**. Translating this sentence into symbol → $C = C^T$
    2. It is **Nonnegative Definite:** Translating this sentence into symbol → $\underline{\boldsymbol{a}}^T \mathrm{C}\, \boldsymbol{a} \geq 0 \quad \forall \boldsymbol{a}$
    3. $\det(\mathrm{C}) \geq 0$
  - **Definition**. correlation matrix of the random vector **x** is equal to

$$\mathrm{R}_{\underline{\mathbf{x}}} = \mathrm{E}(\underline{\mathbf{x}} \cdot \underline{\mathbf{x}}^{\mathrm{T}})$$

Sequences of Random Variables

- **Expectation, Covariance and Correlation**
  - More generally, we can define the following matrices**:**

$$C_{\underline{x}\underline{y}} = \mathrm{cov}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = \mathrm{E}\left[(\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}})(\underline{\mathbf{y}} - \eta_{\underline{\mathbf{y}}})^{\mathrm{T}}\right] \quad : \text{Cross Covariance}$$

$$R_{\underline{x}\underline{y}} = \mathrm{E}(\underline{\mathbf{x}} \cdot \underline{\mathbf{y}}^{\mathrm{T}}) \quad : \text{Cross Correlation}$$

  - We call the $\mathbf{x}_i$s **uncorrelated** if for each j ≠ i we have: $\sigma_{ij} = 0$ i.e. $C_{\underline{\mathbf{x}}}$ is diagonal.
  - We call $\mathbf{x}_i$s **orthogonal**, if for each j ≠ i we have: $E(\mathbf{x}_i \mathbf{x}_j) = 0$ i.e. $R_{\underline{\mathbf{x}}}$ is diagonal.
  - Similar to the two-dimensional mode(two random variables), it can be shown that, if the $\mathbf{x}_i$s are **independent**, they will also be uncorrelated. Also, $g_i(\mathbf{x}_i)$s will be uncorrelated.

Sequences of Random Variables

- **Conditional <u>Distributions</u>**
  - Similar to the two-dimensional mode(two random variables), we have

$$f_{\mathbf{x}_1\mathbf{x}_2\cdots\mathbf{x}_k}(x_1,x_2,\ldots,x_k\,|\,\mathbf{x}_{k+1}=x_{k+1},\ldots,\mathbf{x}_n=x_n)=\frac{f_{\mathbf{x}_1\mathbf{x}_2\cdots\mathbf{x}_n}(x_1,x_2,\ldots,x_n)}{f_{\mathbf{x}_{k+1}\cdots\mathbf{x}_n}(x_{k+1},\ldots,x_n)}$$

  - For example:

$$f_{\mathbf{x}_1\mathbf{x}_2}(x_1,x_2\,|\,x_3,x_4)=\frac{f_{\mathbf{x}_1\mathbf{x}_2\mathbf{x}_3\mathbf{x}_4}(x_1,x_2,x_3,x_4)}{f_{\mathbf{x}_3\mathbf{x}_4}(x_3,x_4)}$$

  - Using the first relationship, it can be shown that:

$$f_{\mathbf{x}_1\mathbf{x}_2\cdots\mathbf{x}_n}(x_1,x_2,\ldots,x_n)=f_{\mathbf{x}_1}(x_1)f_{\mathbf{x}_2}(x_2\,|\,x_1)f_{\mathbf{x}_3}(x_3\,|\,x_2,x_1)\cdots f_{\mathbf{x}_n}(x_n\,|\,x_1,x_2,\ldots,x_{n-1})$$

Sequences of Random Variables

- **Conditional Expectation**

$$E(\mathbf{y}|\mathbf{x}_1 = x_1, \mathbf{x}_2 = x_2, \ldots, \mathbf{x}_n = x_n) = E(\mathbf{y}|x_1, x_2, \ldots, x_n) = \int_{-\infty}^{+\infty} y\, f_{\mathbf{y}}(y|x_1, x_2, \ldots, x_n)\, dy$$

$$E(g(\mathbf{y})|x_1, x_2, \ldots, x_n) = \int_{-\infty}^{+\infty} g(y)\, f_{\mathbf{y}}(y|x_1, x_2, \ldots, x_n)\, dy$$

- **Properties:**

1. $E[E(\mathbf{y}|\underline{\mathbf{x}})] = E(\mathbf{y})$
2. If $\underline{\mathbf{x}}$ and $\mathbf{y}$ are independent, then $E(\mathbf{y}|\underline{\mathbf{x}}) = E(\mathbf{y})$

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**

1. Random Sums

- Given a discrete type random variable **N** (with mean η_N and σ_N^2)taking the values 1, 2, ... and a sequence of random variables $\mathbf{x}_i$ independent of **N**, we form the sum

- and

$$\mathbf{y} = \sum_{i=1}^{\mathbf{N}} \mathbf{x}_i$$

$$\forall i : E(\mathbf{x}_i) = \eta_{\mathbf{x}} \, , \, \mathrm{var}(\mathbf{x}_i) = \sigma_{\mathbf{x}}^2$$

- For example, **y** can be the daily profit of a shop, $\mathbf{x}_i$ is the profit of each customer, and **N** is the number of customers who visit per day.

- Mean of **y** ?
  a) **Intuitively**: ….
  b) **Mathematically**

$$E(\mathbf{y}) = E[E(\mathbf{y}|\mathbf{N})]$$

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**

1. **Random Sums**

$$E(\mathbf{y}|\mathbf{N}=n)=E\left(\sum_{i=1}^{n}(\mathbf{x}_i|n)\right)=\sum_{i=1}^{n}E(\mathbf{x}_i|n)=\sum_{i=1}^{n}E(\mathbf{x}_i)$$

$$\forall i: E(\mathbf{x}_i)=\eta_{\mathbf{x}}$$

$$E(\mathbf{y}|n)=n\eta_{\mathbf{x}}$$

$$\Rightarrow E(\mathbf{y}|\mathbf{N})=\mathbf{N}\eta_{\mathbf{x}}$$

$$\Rightarrow E(\mathbf{y})=E\left[E(\mathbf{y}|\mathbf{N})\right]=E(\mathbf{N}\eta_{\mathbf{x}})=\eta_{\mathbf{x}}E(\mathbf{N})=\eta_{\mathbf{x}}\eta_{\mathbf{N}}$$

Probability theory works

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- Consider the random variation of **x** to be **height of men** and **assume** (based on arguments or long observation) that it has a probability density function of $f_{\mathbf{x}}$ (e.g., N(170, 10)).

- Now let's take a man as an **sample** (completely randomly) and ask you what is the probability that this man is smaller than x in height?

$$P\{\mathbf{x}_1 \le x\} = F_{\mathbf{x}}(x)$$

  - Similarly, for other samples, we have:

$$F_{\mathbf{x}_i}(x) = F_{\mathbf{x}}(x)\,,\; f_{\mathbf{x}_i}(x) = f_{\mathbf{x}}(x)$$

  - On the other hand, assuming the **independence of the experiments**, the x_is are independent.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- Or for example, if the random variable **x** is **monthly data usage** (in gigabytes) of customers using a mobile data plan and suppose it has a pdf of $f_{\mathbf{x}}$, this is a model that is true for all **x** s according to the assumption, so if we denote $\mathbf{x}_i$ as the data usage of the i-th customer,, we have

$$f_{\mathbf{x}_i}(x) = f_{\mathbf{x}}(x)$$

- and the $\mathbf{x}_i$s (assuming the independence of the experiments) are independent.

- In general, random variables that are independent and have the same distribution are called **i.i.d. (Independent Identically Distributed).**

- If $\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$ and the $\mathbf{x}_i$s are i.i.d. we have

$$f_{\underline{\mathbf{x}}}(\underline{x}) = f_{\mathbf{x}_1}(x_1)\cdots f_{\mathbf{x}_n}(x_n) = f_{\mathbf{x}}(x_1)\cdots f_{\mathbf{x}}(x_n)$$

Independence (↓ over $=$) Identically Distributed (↓ over $=$)

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- By repeatedly sampling a random variable in a **population** (e.g., the population of customers in the example above), n i.i.d. random variables are obtained such that:

The entire sample can be represented as a sequence of random variables.

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

$$f_{x_i}(x) = f_x(x)$$

$$\Rightarrow E(x_i) = E(x) = \eta$$

$$\Rightarrow var(x_i) = var(x) = \sigma^2$$

- n is called a **sample size**.
- **Independence**: Each random variable does not influence or provide information about other random variable
- **Identically Distributed**: All observations come from the same probability distribution. The underlying statistical properties (like mean and variance) remain constant across all observations.
- Sampling plays an essential role in statistics.

Sequences of Random Variables



- **Some applications of sequence of random variables and conditional expectation**

2. Sampling

- An i.i.d. sample is often called a ***random sample*** in the statistics literature.
- Note that the word "**sample**" can refer to both a collection of random variables and to a single random variable. It should be clear from the context which meaning is being used.
- **Sample Statistics**: Functions of random samples (e.g., **sample mean**, **sample variance**) are themselves **random variables** (when considered across all possible samples).
- **Practical Implications**

1. **Estimation**: Random samples allow us to estimate **population parameters** using **sample statistics.**
2. **Inference**: The properties of random samples form the basis for **statistical inference**.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**

2. Sampling



- **Practical Implications**

3. **Law of Large Numbers**: As the sample size(n) increases, sample statistics tend to converge to their corresponding population parameters (Note that sample statistics are random variables).

4. **Central Limit Theorem:** For large sample sizes, the distribution of sample mean approaches a normal distribution, regardless of the underlying population distribution (Note that sample mean is a random variables).

- In summary, while a random variable represents a single uncertain quantity, a random sample is a collection of such quantities drawn from the same population. This collection of random variables (the random sample) allows us to make inferences about the population from which they are drawn.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- When we draw a sample from a population: each **observation** is randomly selected from the population and prior to selection, we don't know which specific value from the population will be chosen. Therefore, each observation in the sample is itself a random variable

- **Realization vs. Random Variable**

  - It's important to distinguish between:

    1. The **observed value (realization)** after sampling, which is a fixed number.

    2. The random variable before sampling, which represents the potential outcomes.

- Treating each observation as a random variable, allows us to use probability theory.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- In conclusion, each observation in a sample is considered a random variable because it inherits the randomness and probabilistic nature of the sampling process. This concept is crucial for understanding **sampling distributions** and making **statistical inferences** about populations based on sample data.

- **Question:** Is a random sample a collection of actual observed values or a collection of random variables?

  - A random sample can be viewed from two perspectives:

    1. **Before Sampling (Theoretical Perspective)**

    2. **After Sampling (Observed Perspective)**

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Before Sampling: Collection of Random Variables**

Before the sampling process occurs:

- A random sample is conceptualized as a collection of random variables.

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

- Each element in the sample, $\mathbf{x}_1$, $\mathbf{x}_2$, ..., $\mathbf{x}_n$, is a random variable.
- These random variables represent the potential outcomes of the sampling process.
- These random variables are i.i.d. and each variable has a probability distribution identical to the population distribution.
- We can analyze theoretical properties like expected values and variances.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **After Sampling: Collection of Observed Values**

Once the sampling process is completed:

$$\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

- The random sample becomes a collection of actual observed values.
- These are specific realizations of the random variables.
- Each observed value is a fixed number drawn from the population.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Example**

  Consider flipping a fair coin 3 times:

  - **Before flipping (Random Variables):** $\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x_1} \\ \mathbf{x_2} \\ \mathbf{x_3} \end{bmatrix}$, where each $\mathbf{x}_i$ can be either Heads (1) or Tails (0) with probability 0.5.

  - **After flipping (Observed Values):** We might get $\underline{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, representing Heads, Tails, Heads.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**

2. Sampling



- Understanding both perspectives is crucial.
- The theoretical view (random variables) allows us to create probabilistic models and develop statistical methods, while the practical view (observed values) enables us to apply these methods to real-world data.
- **Importance of i.i.d. Assumption**

  The assumption that observations are independent and identically distributed (i.i.d.) is fundamental in many statistical and probabilistic analyses.

  1) **Simplifies Analysis:** It allows for the application of many statistical theorems and techniques.

  2) **Enables Inference**: It supports drawing conclusions about a population from a sample.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Importance of i.i.d. Assumption**

3) Supports Key Theorems:
- The **Law of Large Numbers** relies on the i.i.d. assumption.
- The **Central Limit Theorem** is based on i.i.d. random variables.

4) Ensures Representativeness: i.i.d. samples are more likely to be representative of the population.

5) Facilitates Model Building: Many statistical models and machine learning algorithms assume i.i.d. data.

6) Allows for Probability Calculations: The joint pdf of i.i.d. random variables is the product of their individual pdfs.

7) Supports Maximum Likelihood Estimation: The i.i.d. assumption simplifies the likelihood function in statistical inference.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Definition: Sample Mean**

  - For i.i.d. random variables $x_1, x_2, \ldots, x_n$, the **sample mean**, denoted by $\bar{\mathbf{x}}$, is defined as

$$\bar{\mathbf{x}} \triangleq \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$$

  - we have

$$E(\bar{x}) = \frac{1}{n}\sum_{i=1}^{n} E(x_i) = \frac{1}{n}\cdot n\eta = \eta$$

$$\sigma_{\bar{x}}^2 = \frac{1}{n^2}\sum_{i=1}^{n} \sigma_{x_i}^2 = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}$$

Note that:
1. Population Mean: η
2. Population Variance: σ^2

  - Thus, the higher n is, the closer the value of $\bar{\mathbf{x}}$ will be to the real η. $\bar{\mathbf{x}}$ is **sample mean**, while η is the **mean(expectation) of the population**(assumed model parameter).

  - **Question:** Why is $\bar{\mathbf{x}}$ a random variable?

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Definition: Sample Mean**

Population Parameter $\eta=10$

$$\underline{x}=\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$\overline{x} \triangleq \frac{1}{n}\sum_{i=1}^{n} x_i$$

Population

Sample 1 → $\overline{x}_1 = 12$

Sample 2 → $\overline{x}_2 = 13$

Sample z → $\overline{x}_3 = 15$

This statistic is called the sample mean. The distribution of these means, or averages, is called the "sampling distribution of the sample mean". This distribution is normal $N(\eta, \sigma^2/n)$

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Definition**. In the context of statistics, function of one or more random variables(constituting a random sample) that do not depend on an unknown parameter is called a **statistic.**
- In sampling, a function of the vector of random variables of the sample $\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$ is called a statistic.
- **Examples of Statistic: sample mean** and **sample variance**.
- When a statistic is used for **estimating** a population parameter, it is called an **estimator**.
- The sample mean is an **unbiased estimator** of the **population mean**.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Sampling Distribution**
  - The concept of a sample as a random variable becomes even clearer when we consider the **sampling distribution**:

    - If we repeatedly take samples of the same size from a population, each sample will likely yield different values for **sample statistics** like the **sample mean** and **sample variance**.

    - Thus, sample statistics calculated from random samples become **random variables** themselves when considered across all possible samples.

    - The probability distribution of **these sample statistics** across all possible samples is called the **sampling distribution**.
    - In many contexts, only one sample is observed, but the sampling distribution can be found theoretically.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Sampling Distribution**

  - For large sample sizes, sampling distributions of many statistics (e.g., sample mean) tend to follow **normal distribution** (according to **Central Limit Theorem**).

  - The **standard deviation** of a sampling distribution is called the **standard error**. It quantifies the variability of the sample statistic across different samples.

  - Sampling distribution form the basis for **statistical inference**, allowing us to make probability statements about **population parameters** based on **sample statistics**.

  - Larger sample sizes generally lead to sampling distributions with smaller standard errors, meaning the sample statistic varies less from sample to sample.

Sequences of Random Variables

- **Some applications of sequence of random variables and conditional expectation**



2. Sampling

- **Sampling Distribution**

  - Sampling distributions are important in statistics because they provide a major simplification en route to **statistical inference**.

  - More specifically, they allow analytical considerations to be based on the **probability distribution of a statistic**, rather than on the **joint probability distribution** of all the individual sample values.

  - **Summary**:
    The sampling distribution of a statistic is the distribution of that statistic, considered as a random variable, when derived from a random sample of size n.

Sequences of Random Variables

❑ Central Limit Theorem(CLT)

- Two main results in probability are the **law of large numbers** and the **central limit theorem**. Both are limit theorems involving sums of independent random variables.

- In particular, consider a sequence $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ of iid random variables with finite expectation η and finite variance σ^2 and

$$\overline{\mathbf{x}} \triangleq \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$$

- The *Central Limit Theorem* describes the approximate distribution of $\overline{\mathbf{x}}$, and it applies to both continuous and discrete random variables. Loosely, it states that

the average of a large number of iid random variables
approximately has a normal distribution.

- Specifically, when $n \to \infty$ the random variable $\overline{\mathbf{x}}$ has a distribution that is approximately normal, with expectation η and variance $\frac{\sigma^2}{n}$: $\overline{\mathbf{x}} \sim N(\eta, \frac{\sigma^2}{n})$

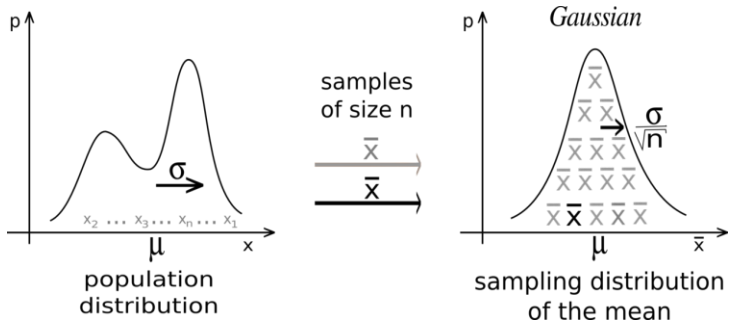
Sequences of Random Variables

❑ Central Limit Theorem

- Thus, the limit as n $\rightarrow \infty$ of the distribution of $\mathbf{z} = \frac{\bar{\mathbf{x}}-\eta}{\frac{\sigma}{\sqrt{n}}}$, is a normal distribution with mean 0 and variance 1.

- According to CLT, it becomes clear why many phenomena in the real world have an **almost normal** distribution. Basically, whenever a phenomenon is influenced by many random factors (a person's height, measurement error, thermal noise voltage, etc.), it will be almost normal.

- Whatever the form of the population distribution, the sampling distribution of the sample mean tends to a Gaussian(normal) distribution.



Sequences of Random Variables

❑ Law of Large Numbers

- Consider a sequence $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ of iid random variables with finite expectation η and finite variance σ^2 and
- We have

$$\bar{\mathbf{x}} \triangleq \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$$

- In other symbols

$$P\{|\bar{x}_n - \eta| > \varepsilon\} \xrightarrow[n\to+\infty]{} 0$$

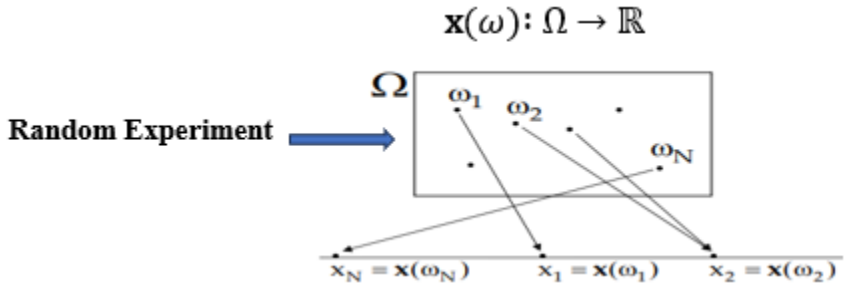
$$\bar{x}_n \xrightarrow[n\to+\infty]{P} \eta$$

- We know $E(\bar{x}) = \eta \quad , \quad \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$

Stochastic Process or Random Process

- **General Concepts**

  - As we recall, a random variable **x** is a rule for assigning to every outcome ω of an experiment S a number **x**(ω).



  - **Definition:** A **stochastic process** **x**(t) is a rule for assigning to every ω a **function** **x**(t, ω).

  - Thus a stochastic process is a family of time functions depending on the parameter ω or, equivalently, a function of t and ω.

  - The domain of ω is the set of all experimental outcomes and the domain of t is a set $\mathbb{R}$ of real numbers.

Stochastic Process or Random Process

- **General Concepts**

  - If $\mathbb{R}$ is the real axis, then $\mathbf{x}(t)$ is a **continuous-time** process.

  - If $\mathbb{R}$ is the set of integers, then $\mathbf{x}(t)$ is a **discrete-time** process.

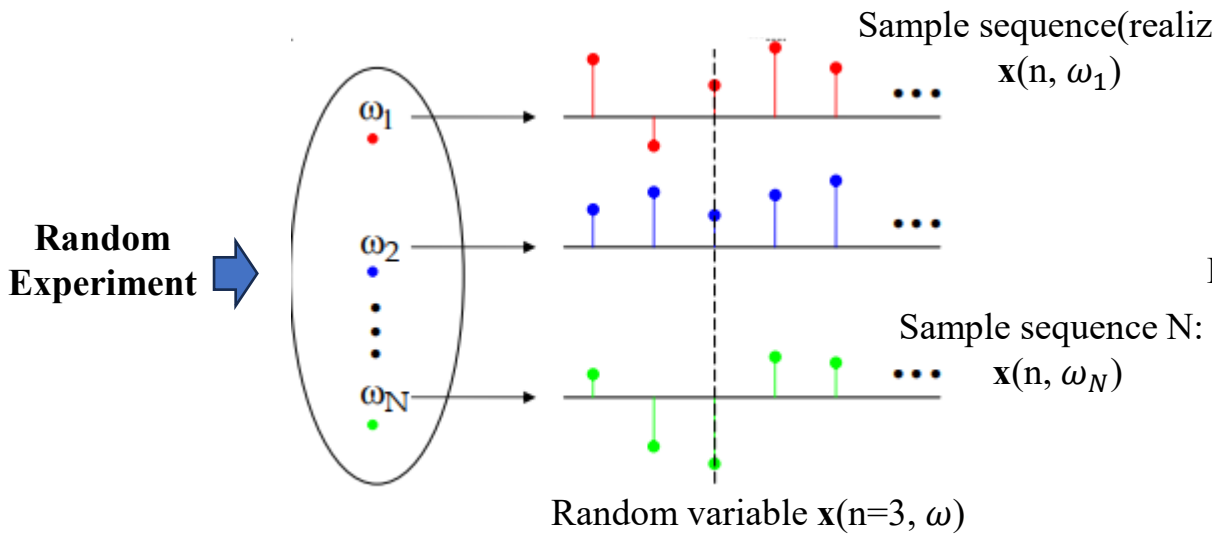
  - A discrete-time process is, thus, a sequence of random variables. Such a sequence is denoted by $\mathbf{x}_n$.

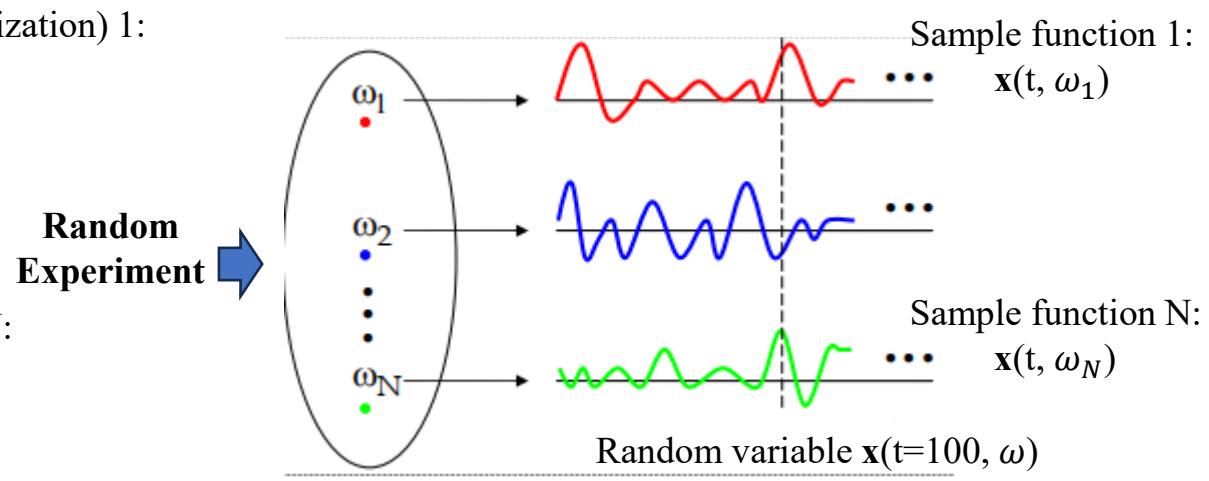
  - We use the notation $\mathbf{x}(t)$ to represent a stochastic process omitting its dependence on ω (as in the case of random variables)

  - $\mathbf{x}(t)$ has the following interpretations:
    1) It is a family (or an ensemble) of functions $\mathbf{x}(t, \omega)$. In this interpretation, t and ω are variables.
    2) It is a single time function (or a sample of the given process). In this case, t is a variable and ω is fixed.
    3) If t is fixed and ω is variable, then $\mathbf{x}(t)$ is a random variable equal to the state of the given process at time t.
    4) If t and ω are fixed, then $\mathbf{x}(t)$ is a number.

Stochastic Process or Random Process

- **General Concepts**
  - Discrete-Time Process **x**(n, ω)
  - Continuous-Time process **x**(t, ω)





- A stochastic process is a noncountable infinity of random variables, one for each t. For a specific *t*, **x**(t) is a random variable with distribution

$$F(x,t) = P\{\mathbf{x}(t) \le x\} \qquad f(x,t) = \frac{\partial F(x,t)}{\partial x} \qquad \dots$$

Stochastic Processes Course

➢ **A summary of the mathematical objects we learned**

Event

Random Variable

- One Random Variable
- Two Random Variables
- Sequence of Random Variables

Example for Sequence of Random Variables: Random Sample

Sample Statistic

- Sample Mean
- Sample Variance

Sampling Distribution

Stochastic Process

- Now, By these new words(objects), We can think better, **model** different phenomena and **solve problems**.

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